

Quantification

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1. Introduction

The aim of this review chapter is to provide an introduction to the formal diachronic study of the semantics and syntax of quantification. This simple definition contains certain important restrictions which need to be elucidated before we proceed to the discussion proper.

By **diachronic** study, I mean works which describe how the system of quantification has changed over time in a given language or a set of languages. This excludes work which merely explores a snapshot of the system of quantification at a given point of time (be it in the present or the past). By **formal** study I mean approaches which adopt or at least are informed by some formal frameworks of syntax or semantics and thus, aspire to move beyond the level of philological description and typological generalisation.

The term **quantification** also needs clarification: while for most linguists, the prototypical case of quantification is universal quantification over a set of individuals in the sense of Barwise & Cooper (1981), the landscape is very broad and diverse (cf. Bende-Farkas 2024 for an overview). For reasons of space, and also in order to maintain a sensible division of labour with other chapters, I will focus on phenomena which at some level involve universal quantification in the logical sense. This decision does not result in a significant loss of coverage: the bulk of formal diachronic study of quantification has indeed been devoted to universal quantification; while the study of existential quantification has been, with good reason, subsumed under the study of indefinites. Also, since diachronic change is often the result of reinterpretation through the mechanism of constant entailments (different constructions having similar or identical truth conditions), our discussion will also touch upon related phenomena such as maximality operators, pluractionals, free-choice items (and indeed, indefinites).

1.1 Emerging themes and strands of research

The formal diachronic study of quantification is in the first stages of rapid development, powered by the advent of large and well-annotated electronic corpora and the advances of formal linguistics. Already, some cross-linguistic generalizations are emerging.

On the one hand, general principles of language change have been shown to be at work when it comes to the diachrony of quantification in particular. This includes the **Head Preference Principle** (van Gelderen 2004), which describes the tendency of phrases to be reinterpreted as heads, but not vice versa. This principle is reflected in the diachronic change of quantification as well, as evidenced by the work of Haspelmath (1995), Martzloff (2014), Bende-Farkas (2014a) and Xing (2015) a.o. **Grammaticalization as the semanticization of pragmatically inferred information** has been proposed as another general driver of language change by Eckardt (2006). This mechanism (or the related concept of ‘pragmatic strengthening’ due to König and Traugott 1988) has been evoked in the analysis of the diachrony of quantification by Haspelmath (1995), Bende-Farkas (2014a), Beck (2017) and Doron (2020).

On the other hand, grammaticalization (or reinterpretation) pathways (cf. WBCDL025 in this volume) specifically related to the diachrony of quantification have been identified. This includes

the **whole-to-all cycle** or spiral: the cross-linguistic tendency of adjectives with the meaning of ‘healthy, whole, intact’ to be reinterpreted as ‘all’-type quantifiers. This tendency has been described in the seminal work of Haspelmath (1995) and has spurred much research since including Mladenova (2009), Martzloff (2014), Xing (2015), Bende-Farkas (2020) and Doron (2020).

The **reinterpretation of adverbial quantifiers (A-quantifiers) as determiner quantifiers (D-quantifiers)** is also widely attested. Phenomena under this rubric are discussed by Haspelmath (1995), Mladenova (2009), Xing (2015) and Bende-Farkas (2020). Since so-called pluractional adverbials (e.g. *day by day*) have truth conditions which are often very close to those of universal quantifiers (e.g. *every day*), **pluractionals are often reinterpreted as universal quantifiers**. Such phenomena are discussed by Haspelmath (1995), Mladenova (2009) and Bende-Farkas (2014a, 2015).

The **Universal Semantic Cycle** has been proposed by Beck (2017, 2020), inspired by an observation of Haspelmath (1995): it describes a four-stage cyclical change from (i) covert quantification of propositional alternatives to (ii) covert quantification over lexical alternatives to (iii) quantification in the sense of generalized quantifiers (Barwise & Cooper 1981) to (iv) loss of quantification. Work in this very promising paradigm includes: Beck (2017, 2020, 2022) and Doron (2020). Bende-Farkas (2014a, 2015) to some extent anticipates parts of Beck’s (2017) proposal.

The mechanism of **ontological semantic functionalization** has been described by Gergel (2014) in connection with the diachrony of *most*: this is a change where at the outset, lexical item A and lexical item B combine compositionally and after reinterpretation, the original semantic contribution of item B is incorporated into the lexical semantics of item A.

1.2 Challenges ahead and open questions

The formal study of quantification has mostly focused on semantics: there is a probably much to explore when it comes to **syntax and the syntax-semantics interface**. Interesting work has been done on this with regard to Hungarian, where it has been shown that the SOV to SVO change and the resulting development of an articulated functional left periphery has led to the development of a dedicated quantifier field and the syntactically overt marking of quantifier scope (É. Kiss 2014b). Since the SOV to SVO change has affected hundreds of languages, this appears a promising line of research.

In terms of semantics: as we will see, type-theoretical **formal semantics** has the potential to be a powerful tool in the description and analysis of quantificational systems undergoing change. There is some promising work on this (Gergel 2014, Bende-Farkas 2014a, 2015, Beck 2017, 2020, 2022, Doron 2020 a.o.), but there is a considerable amount of empirical data, such as the whole-all spiral, to which such tools have not yet been applied systematically.

2. Setting the stage – The sources of all and every

2.1 Haspelmath (1995)

In his seminal paper, Haspelmath (1995) investigated the diachronic sources of “universally quantifying determiners corresponding to English *all* and *every*”. Haspelmath delineates the two subtypes using surface syntactic criteria: *all*-type universally quantifying determiners (UQDs) associate with plural count nouns and mass nouns, whereas *every*-type UQDs associate with

singular nouns. (Note that this syntactic pattern is closely correlated to semantic type: *every*-DPs are usually analysed as distributive and *all*-DPs as collective.)

Haspelmath's main claim is that the current syntactic distribution (and semantic interpretation) of these types of UQDs is a reflection of their respective origins: *all*-type UQDs generally derive from an adjective meaning 'whole'; whereas *every*-type UQDs have three common sources: (i) free-choice indefinite determiners such as English *any*, (ii) distributive [in current terminology, pluractional] prepositions and (iii) *all*-type UQDs.

Haspelmath covers a broad range of languages and language families, although the depth of empirical discussion is uneven and the focus is mainly on Indo-European.

The *whole* → *all* change is well attested in Romance and Greek: while Latin *totus* meant 'whole', its Romance derivatives (such as Portuguese *todo*) have a split interpretation: with singular count nouns, they retain the old meaning of 'whole' and with plural count nouns, they have the interpretation 'all'.

- (1) a. *toda a casa* 'the whole house'
- b. *todos as amigas* 'all the friends'

There has been a very similar change from Ancient Greek *hólos* 'whole' to Modern Greek *ólos* 'all'; and there is an ongoing process in Modern German, where *gan̩z* retains its original meaning of 'whole' alongside with an emerging more innovative 'all'-use in the colloquial register. A similar process has been reconstructed concerning Sanskrit *sarva-* 'whole, all' (<PIE **solmos* 'sound, well'). Hindi-Urdu *saara* 'all' developed from Sanskrit *sāra-* (originally 'strong, firm', reinterpreted as 'sound, whole'), Germanic *all* (from **allo* 'grown up', later reinterpreted as 'complete, whole'), and Greek *pās* (cf. Sanskrit *śvā-* 'swell'). Later work has found similar phenomena in other languages such as Bulgarian (Mladenova 2009), Chinese (Xing 2015) and Hungarian (Bende-Farkas 2020).

Haspelmath hypothesises that the reinterpretation of 'whole'-words as 'all' happens gradually along the following cline (examples are from colloquial spoken German):

- (2) a. singular count nouns (*die ganze Tasse* 'the whole teacup')
- b. collective nouns (*die ganze Familie* 'the whole family') and mass nouns (*das ganze Wasser* 'all the water')
- c. aggregates (i.e., plural individuals in the sense of Link (1983)): *die ganzen Tassen* 'all the teacups'

In later work on Bulgarian, Mladenova (2009) proposes and empirically supports a more nuanced cline (see later, together with a detailed evaluation of various similar proposals).

Haspelmath also discusses instances where "words meaning 'all' are not derived from words meaning 'whole'." This includes the postpositional quantifier *-tēr-* (followed by person/number suffixes) in Egyptian/Coptic, which derives from the postposed prepositional phrase *r d̩r* (+person/number) 'to X's end'. Haspelmath further shows that there was an intermediate state in Ancient Egyptian where *r d̩r* was an adverbial universal quantifier, before it solidified as an adnominal quantifier; making this an instance of an A-quantifier being reanalysed as a D-quantifier.

Turning to the sources of ‘every’, Haspelmath points out that in many languages, “the universal distributive determiner ‘every’ is synchronically or etymologically derived from a wh-determiner [a wh-indeterminate or indeterminate pronoun in current terminology] by means of a special particle (originally meaning ‘also’, ‘even’, ‘or’, ‘it may be’ etc.).” In some languages, this is still visible on the surface:

- | | | | |
|----------------|---------------|--------------|--------------|
| (3) a. Chinese | <i>shéi</i> | <i>ye</i> | ‘everyone’ |
| | who | also | |
| b. Korean | <i>nwukwu</i> | <i>-na</i> | ‘everyone’ |
| | who | or | |
| c. Rumanian | <i>fie</i> | <i>-care</i> | ‘every(one)’ |
| | be.SUBJ.3SG | who/which | |

In others, it has been obscured through morphophonological reduction: e.g. Modern English *each* (< OE *ælc* < Proto-Germanic **ajw-hwalik* ‘ever-which’).

Furthermore, Haspelmath points out that in an even greater range of languages, free-choice items are composed similarly (Bremen 1983, Haspelmath 1997):

- | | | | |
|--------------|-------------|-----------------|-------|
| (4) a. Latin | <i>qui</i> | <i>-vis</i> | ‘any’ |
| | which | you want | |
| b. Polish | <i>jaki</i> | <i>-kolwiek</i> | ‘any’ |
| | which | ever | |
| c. Basque | <i>zein</i> | <i>-nahi</i> | ‘any’ |
| | which | want | |

Haspelmath argues that the reinterpretation happened in two steps: some element + wh-determiner -> free-choice item -> universal.

Haspelmath identifies two sources for such paradigms: non-specific free relative clauses and parametric concessive conditional clauses (unconditionals is current terminology). In the first case, free relative clauses containing a wh-determiner and some other element (concessive particle, additive particle, *want*, *it may be* etc.) were reanalysed in the following way: the ‘other element’ was reinterpreted as an ‘indefiniteness marker’ and it morphologically combined with the wh-determiner, yielding a free-choice item:

- | | | | | | | | |
|--------|------------|-------------|--------------|------------------|---------------|--------------|--------------|
| (5) a. | <i>Ela</i> | <i>pode</i> | <i>tomar</i> | <i>qual</i> | <i>coisa</i> | <i>quer.</i> | (Portuguese) |
| | She | can | take | what | thing | she wants. | |
| -> b. | <i>Ela</i> | <i>pode</i> | <i>tomar</i> | <i>qual-quer</i> | <i>coisa.</i> | | |
| | She | can | take | any | thing. | | |

With parametric concessive conditionals, the picture is similar: the antecedent of the conditional contains a wh-indeterminate and some other element (concessive, *be* etc.), and the two are at some point reinterpreted and morphologically combined to yield a free-choice item:

- (6) a. *Where* *ever* *she goes, I will never leave her.*

wh-indeterminate concessive particle

- b. *Wherever* *she goes, I will never leave her.*

free choice item

More examples and descriptive details are offered in Haspelmath (1997). Formal semantic accounts for this process are offered in later work by Beck (1997, 2020, 2022) and Halm (2023).

The second step is the reinterpretation of the free-choice item as a universal quantifier. Haspelmath argues that due to their well-known quasi-universality, free-choice items and universals very often have nearly identical truth conditions:

- (7) a. *Anyone can solve this puzzle.*
b. *Everyone can solve this puzzle.*ⁱ

As Haspelmath points out, even in situations where there are differences in meaning, the sentence containing the FCI very often conversationally implicates its counterpart with the universal:

- (8) a. *Anyone come come in.* --implicates--> b. *Everyone can come in.*

Thus, Haspelmath argues, the FCI-to-universal reinterpretation is an instance of ‘pragmatic strengthening’ (König and Traugott 1988, see also WBCDL127), or, to use later parlance, of the semanticization of pragmatically inferred material (Eckardt 2006, see also WBCDL120), and it relies on a configuration characterized by constant entailments (Beck 2012, Gergel & Beck 2015): a situation where two compositionally different structures have the same (or very similar) truth conditions.

Another trajectory discussed by Haspelmath is when ‘every’ derives from a ‘distributive preposition’ (a pluractional preposition in current terminology, cf. Beck 2021 and references). A case in point is Ancient Greek *katá*, which in addition to its basic locative meaning ‘along, throughout’ could also be “used to indicate that a place or point in time is distributed over a multiple event [sic]”. As has been pointed out in the literature on pluractionals (Beck 2021, Halm & Bende-Farkas 2024), pluractionality often has roughly the same quantificational import as universal adverbial quantification (day-by-day vs. every day, one-by-one vs each):

- (9) *eporeúnto ho goneĩs autoĩ kat’ étos eis Hierousalē̃m* (Lk 2:41) (Koine Greek)
they.went the parents his pluract year to Jerusalem

‘His parents went to Jerusalem year-by-year (=every year).’

This creates an environment conducive to reinterpretation; and indeed, Haspelmath shows that such a reinterpretation has taken place in Greek. Ancient Greek *kath’ hḗna* ‘(one) by one’ has been reinterpreted already in the Ancient Greek period as *kathḗis* ‘everyone’. In Modern Greek, *kathís* is the colloquial word for ‘everyone’ (having crowded out *pās*), and *kátthe* (which Haspelmath a shortened form of *kathís*) is the determiner ‘every’. This process thus represents a change from (i) pluractionality to universal quantification and from (ii) A-quantification to D-quantification; and it also an instance of (iii) phrase-to-head reinterpretation (as the DP *kathís* ‘everyone’ is

reinterpreted as the D *kathé* ‘every’) (in line with the Head Preference Principle, van Gelderen 2004).

Ancient Greek *katá* is the source (having been borrowed into Late Latin) of Portuguese and Spanish *cada* ‘every’ and possibly (through a blend of Latin *quisque* and Vulgar Latin *cata unum*) of Italian *ciascuno* and French *chacun*, and thus French *chaque* (through back-formation). While Haspelmath does not differentiate in the glosses between ‘every’ and ‘each’, all the examples in this section are better translated as ‘each’, which might be connected to the fact that they derive from pluractionals which are inherently distributive. Haspelmath briefly discusses an analogous development in Indo-Aryan, where Hindi-Urdu *pratyek* ‘every’ is derived from *ek* ‘one’ and the Old Indic pluractional postposition *–prati*; as well as cases where ‘every’ is derived from reduplicated numerals or wh-indeterminates.

Haspelmath also observes that ‘all’ can be source for ‘every’, with the typical pattern that the same lexeme is used in both senses: the ‘all’ reading is elicited in combination with definite plural NPs whereas the ‘every’ reading is elicited in combination with bare singular NPs:

- (10) a. *todas as casas* *toda casa* (Portuguese, representative of Romance)
 ‘all the houses’ ‘every house’
 b. *pántes hoi hodoí* *pāsa hodós* (Modern Greek)
 ‘all the roads’ ‘every road’
 c. *kol ha-sfarim* *kol sefer* (Modern Hebrew)
 ‘all the books’ ‘every book’

Haspelmath (1995) covers a wide range of languages and makes important generalizations. While this paper lacks the formal rigour or the empirical detail of some later work in the field, most of its observations have stood the test of time and it has also served as an inspiration for much further research.

2.1 Universal quantification in Mandarin Chinese: Xing (2015)

Xing (2015) is a corpus-driven, in-depth analysis of the diachronic processes resulting in the emergence of **the universal quantifier *měi* in Mandarin Chinese**. Spanning three millennia (from the eleventh century BC to the twentieth century AD), Xing (2015) identifies four steps of semantic change.

In Step1, the predicative adjective *měi* ‘flourishing’ (possibly a derivation of *měi* ‘tips of bamboo shoots’) was reinterpreted as *měi* ‘plentiful, abundant’. In Step 2, the predicative adjective *měi* ‘flourishing, abundant’ was reinterpreted as the frequency adverbial *měi* ‘often, frequently’. In Step 3, the frequency adverbial *měi* ‘often, frequently’ was reinterpreted as the temporal adverbial quantifier *měi* ‘every time’. In Step 4, the temporal adverbial quantifier *měi* ‘every time’ was reinterpreted as distributive universal quantifying determiner *měi* ‘every’.

- (11) a. 原田每每, 舍其舊而新是謀。 (4th Century BC, Zuo Zhuan)
 yuántián mēi-mēi shě qí jìn ér xīn shì móu
 field **flourish** rid POSS old but new PART gain

‘The (bamboo) field is flourishing, (someone) gets rid of the old and seeks the new.’

- b. 懷和為每懷, 咨才為諷, (3rd Century BC, *Guoyu*)

huái hé néi měi huái, zī cái wéi zōu
mindful compromise is **plenty** in-mind consult talent is consult

‘Thoughtful compromise takes a lot of thought. To consult talent is to seek advice.’

- c. 事, 猶琴瑟也, 每終改調。 (3rd Century BC, Tongxuan Zhenjing)

shì yóu qínsè yě měi zhōng gǎi diào
things like string-music PART **often** end change tune

‘Things like string music often change melody in the end.’

- d. 每有一事, 群臣同聲 (1st Century AD, Hanshu)

měi yǒu yī shì qún chén tóng shēng
A-quant. have one thing crowd official same voice

‘Every time something comes up, all officials agree on it.’

- e. 每律各一, 內卑外高。 (1st Century AD, Hanshu)

měi lǜ gè yī nèi bēi wài gāo
D-quant. rule each one inside humble outside superior

‘Every rule embodies this: humble inside and superior outside.’

Xing (2015) characterizes Steps 1 and 2 as instances of metaphoricalization and Step 3 as metonymization, whereas Step 4 is described as a case of semantic reanalysis (cf. WBCDL031). While Xing (2015) eschews a formal semantic or syntactic analysis, Step 4 can be straightforwardly characterized as an instance of an A-quantifier being reinterpreted as a D-quantifier, which involves a change in both the semantic type and the domain of quantification ($\langle s, \langle s, t \rangle \rangle^i$ to $\langle e, t \rangle \langle \langle e, t \rangle, t \rangle$). In syntactic terms, Step 4 is a case of phrase-to-head reanalysis. We can relate this to van Gelderen’s (2004) Head-Preference Principle: the concept that language learners/users prefer heads to phrases and thus have a tendency to reanalyse phrases as heads in the course of language acquisition.

2.3 Universal quantifiers of time in Germanic: Leuschner (1996)

Leuschner (1996) provides a systematic account of the meaning history of the three ‘ever’-words: *ever* (English), *je* (German) and *ooit* (Dutch). Synchronically, *ever-je-ooit* are NPIs, licensed (i) by (direct or indirect) negation, (ii) in interrogatives, (iii) in the protasis of conditionals and (iv) in comparatives. Leuschner (1996) argues that a quantificational contrast emerges between the existentially interpreted NPIs *ever-je-ooit* and their respective universal temporal quantifier counterparts *always-immer-altijd* under the scope of negation and in questions.ⁱⁱⁱ

- (12) a. *John didn’t ever imagine this outcome.* $\neg > \exists$

- b. *John didn't always imagine this outcome.* $\neg > \forall$
- (13) a. *Did John ever imagine this outcome?* $Q > \exists$
- b. *Did John always imagine this outcome?* $Q > \forall$

German *je* is the direct descendant of **ain* ‘always, ever’ through OHG *io*. English *ever* is a descendant of OE *ǣfre*, which is a compound of *ā* (direct descendant of **ain*) plus either *feore* ‘in life’ or *byre* ‘in the event’. *Oit* is a descendant of Old Frisian **ô* (direct descendant of **ain*) and Old Low Frankonian **giet* or Old Frisian **ieta* ‘still’, cognates of ModE *yet*. Modern German and Modern Dutch *immer* derive from *io*/**eo* + *mêr*/*mere* ‘more’.

Concerning pre-Modern meanings and meaning distinctions, based on the discussion of a handful of illustrative sentences and dictionary entries, Leuschner finds that in Old and Early Middle English, *ā*/*ǣfre* was ambiguous between a universal temporal (‘always’) and a (negative) existential ‘ever’ reading. Leuschner offers no formal semantic or syntactic analysis; however, this quantificational plasticity is strongly reminiscent of the situation discussed by Beck (2017): therefore, an analysis in terms of indeterminate (temporal) pronouns bound by covert operators (universal, existential, negative) looks eminently plausible. In Old and Middle High German, *io*/*ie* displayed the same universal/(negative) existential ambiguity; and there was also *io* *mêr* / *iemer*, which Leuschner analyses as a version of *io*/*ie* marked for future reference.

The situation in Middle English, Middle High German and Middle Dutch is summarized as follows:

(14)	MidE	MHG	MidD
universal (future)	<i>efre</i> (<i>a</i> + <i>in feore</i>)	<i>iemer</i> (<i>ie</i> + <i>mêr</i>)	<i>emmer</i> (<i>*eo</i> + <i>mer</i>)
universal (past)	<i>efre</i> (<i>a</i> + <i>in feore</i>)	<i>ie</i>	<i>ie, ooit</i> (<i>*ô</i> + <i>*ieta</i>)
NPI existential	<i>efre</i> (<i>a</i> + <i>in feore</i>)	<i>ie</i>	<i>ie, ooit</i> (<i>*ô</i> + <i>*ieta</i>)

There are a number of interesting developments here: the reinforcement in English of *ā* ‘always, ever’ through the absorption of the intensifier *in feore* ‘in life’; the reinforcement in Dutch of **ô* through the absorption of the particle **ieta* ‘yet’; and the reanalysis of [*ie* ‘always/ever’ + *mêr* ‘more’] as *iemer* ‘always, +future’. These changes are plausible enough, however, no formal semantic account is offered here (or indeed later in the literature, as far as I know).

Next, Leuschner discusses the changes that led to the current situation:

(15)	ModE	ModG	ModD
universal	<i>always</i>	<i>immer</i>	<i>altijd</i>
NPI existential	<i>ever</i>	<i>je</i>	<i>ooit</i>

One striking development is the breakdown of a distinction between universal-future and universal-past: in English and Dutch, this is due to the adoption of an ‘all’-expression (*always* and

altijd), whereas in German, the loss of the [+future] specification of *immer*. A similar process happened in Dutch too, but *immer/emmer* ‘always’ was then superseded by *altijd* ‘always’. As Leuschner points out, both in Early Modern German and Dutch, we then see the emergence of *immermehr* ‘always [+future]’ (<*immer+mehr*) and *immermeer* ‘always [+future]’ (<*immer+meer*): a very clear instance of a spiral, where successive cycles of loss and reinforcement involve the same element (*mehr/meer*):

(16) *ie* ‘always’ -> *iemer* ‘always [+future]’ -> *immer* ‘always’ -> *immermehr* ‘always [+future]’

The other major development is the loss of the universal/(negative) existential ambiguity. In light of later developments in the field (specifically, the Universal Semantic Cycle of Beck), one possible analysis that offers itself is that the change involved a transition from a system of ‘covert operators and indeterminate pronouns’ to (adverbial) quantifier phrases in the Barwise & Cooper (1981) tradition.

In later work, Hoekstra, Slofstra & Versloot (2009) augmented the empirical map provided by Leuschner (1996) with a discussion of the diachrony of ‘ever’-words in Frisian. A formal diachronic semantic and syntactic analysis of Germanic ‘ever’-words is, for the time being, outstanding.

2.4 From ‘whole’ to ‘all’ (and back) in Latin and Romance: Martzloff (2014)

Martzloff (2014) offers a detailed exploration of the prehistory of *omnis* and other Latin pronouns and adjectives meaning ‘all’ or ‘whole’. Martzloff characterizes *omnis* as a “simple, unmarked” universal quantifier which could be interpreted distributively as well as non-distributively (collectively). *Omnis* associated both with plural count nouns and singular count nouns. In the latter case, it combined with collective nouns or mass nouns or the names of unique entities such as regions and had a meaning corresponding ‘the whole’. Martzloff shows that telling apart the contribution of *omnis* and *totus* is not trivial, however, both *totus* and *omnis* could associate with mass nouns as well as collective nouns, with similar if not identical meaning. Combining this with the situation in Romance (cf. Haspelmath 1995), the following picture emerges:

(17)	singular count nouns	mass nouns	collective nouns	plural count nouns
Latin	<i>totus</i>	<i>totus/omnis</i>	<i>totus/omnis</i>	<i>omnis</i>
Portuguese	<i>todo</i>	<i>todo</i>	<i>todo</i>	<i>todo</i>

This is consistent with a diachronic process whereby *totus* (originally meaning ‘whole’ and being limited to singular count nouns) gradually spreads to nominals denoting various kinds of pluralities (mass nouns, collective nouns, plural count nouns). As we will see later on, Mladenova (2009) charts an almost identical development in the history of Bulgarian.

Turning to the prehistory of *omnis* itself, Martzloff argues, in agreement with Pauli (1885), Brugmann (1984) and Hackstein (2010), that *omnis* derives from *ops* ‘abundance’ (and not *amplus*, Balles 2009). In addition to phonological and morphological considerations, he also lists numerous parallels from various languages where ‘whole’ or ‘all’ derives from an adjective with the rough meaning of ‘large’.

Finally, Martzloff remarks upon the fact that *omnis* can be used both as a determiner (D head in generative parlance) and as a ‘substantive’ (i.e., as a DP phrase). He tentatively argues that the phrasal use might be the original one, in which case, we have a phrase-to-head reanalysis at hand,

in line with the Head Preference Principle (van Gelderen 2004) and also with the tendency for higher types (von Stechow 1995).

2.5 Quantification in Bulgarian: Mladenova (2009)

In an empirically rich and insightful paper, Mladenova charts the diachronic changes in the system of (mostly) universal quantification in Bulgarian. The main focus is a set of cyclical changes between ‘all’- and ‘whole’-words. As far as Modern Bulgarian is concerned, Mladenova argues that *vsički* ‘all’ and *čjal* ‘whole’ have roughly the same meaning contribution (collective universal quantification), but they pattern differently in terms of the semantic type of their possible associates:

(18)	<i>vsički</i> ‘all’	<i>čjal</i> ‘whole’
a. plural count nouns (<i>jabālki</i> ‘apples’)	✓	✗
b. mass nouns (<i>kiselo mljako</i> ‘yoghurt’)	✓	✗
c. abstract nouns (<i>māka</i> ‘grief’)	(✓)	✓
d. collectives (<i>dobitāk</i> ‘cattle’)	✓	(✓)
e. ‘aggregates’ ^{iv} (<i>narod</i> ‘nation’)	✗	✓
f. singular count nouns (<i>jabālka</i> ‘apple’)	✗	✓

Mladenova adopts, in the vein of Richter’s (1909) and Sapir’s (1930) intuition, a distinction apparently used in Finnish grammar (no sources provided) and also applied to English by Chesterman (1991:164)^v; and she argues that *vsički* ‘all’ is a quantifier of totality which associates with so-called ‘divisible’ nominals whereas *čjal* is a quantifier of totality which associates with ‘indivisible’ nominals.

Next, Mladenova focuses on the dramatic language change that played out between the 17th century and today. While in the 17th century, *čjal* ‘whole’ was only used with singular count nouns, as of today, *čjal* associates with a much larger class of nominals (18). Mladenova argues that this change can be described in a parametric fashion: the semantics of *čjal* ‘whole’ and *vsički* ‘all’ did not change: rather, what changed is which classes of nominals count as divisible or indivisible:

(19) What counts as divisible in Bulgarian?

	17 th C	today
count nouns (singular)	✗	✗
aggregates	✓	✗
collective nouns	✓	✓ (✗)
abstract nouns	✓	✗ (✓)
mass nouns	✓	✓
count nouns (plural)	✓	✓

Mladenova paints a detailed empirical picture of the stages of this change: the transition from divisible to indivisible (i.e., the transition from association with *vsički* ‘all’ to association with *čjal*

‘whole’) first affected aggregates, then abstract nouns, then collective nouns. In the Kostur dialect, this process went even further, and there, apparently all types of nouns now associate with *ĉjal* ‘whole’ only:

- (20) singular count < aggregate < abstract < collective < {mass|plural count}
- | | | | |
|-------------|--------------------|-----------|----------------------|
| WHOLE ----- | ----- | ----- | ---> ALL |
| <i>ĉjal</i> | 17 th C | Modern | Kostur <i>vsički</i> |
| | | Standard | dialect |
| | | Bulgarian | |

We can immediately note that this actually instantiates an implication hierarchy (Mladenova herself does not use this specific term). If, e.g., in any dialect of Bulgarian, collectives associate with *ĉjal* ‘whole’, then it follows that abstract nouns, aggregates and singular count nouns also associate with it. As Mladenova notes, this also raises the intriguing empirical question whether this same implicational hierarchy (or something similar) holds for other languages.

Following Brugmann (1894:60-64), Mladenova points out that in Ancient Greek, *pās* covered the whole space of semantic totality. Then, *bólos* emerged and it gradually crowded out *pās* until *bólos* became the only non-distributive quantifier (Brugman 1894:44-45, Frisk 1970, 2:381). And as of today, in Modern Greek, *holókléros* competes with *bólos* in the case of singular counts nouns and aggregates: this means that the cycle has started again, and the change can be characterized as a spiral:

- (21) (*pās*) -> (*pās* – *bólos*) -> (*bólos*) -> (*bólos* – *holókléros*)

As we have seen, a less fine-grained but rather similar hierarchy of nominal categories emerges in the history of Latin-Romance (Haspelmath 1995, Martzloff 2014):

- (22) singular count < {collective|mass} < plural count
- | | | |
|--------------|-------|----------------------|
| WHOLE ----- | ----- | ---> ALL |
| <i>totus</i> | Latin | Romance <i>omnis</i> |

While in Latin, the space was split between *totus* and *omnis*, in Romance, *totus* covers the whole space. To this, Mladenova adds the observation that in Modern Romanian, while *tot* covers the whole inclusive totality space (and it is also used distributively), *intreg* ‘entire’ (a derivative of Latin *integer* ‘unharméd’) is competing with *tot* as a non-distributive quantifier in the case of singular count nouns, aggregates and collectives; another instance of a spiral:

- (23) (*omnis* – *tot*) -> (*tot*) -> (*tot* – *intreg*)

As we have seen, German *gan̄* is also spreading along a similar cline (Haspelmath 1995):

- (24) singular count < collective < mass < plural count
- | | | |
|-------------|----------|------------|
| WHOLE ----- | ----- | ---> ALL |
| <i>gan̄</i> | Standard | colloquial |
| | Modern | Modern |
| | German | German |

And as we will see later, certain dialects of Hungarian are undergoing a similar change (Bende-Farkas 2020):

(25)	singular count < collective < mass < plural count		
WHOLE	----- ----- ---	> ALL	
<i>egész</i>	Modern	Csángó &	
	Hungarian	Colloquial	
		Transylvanian	
		Hungarian	

The emerging cross-linguistic empirical pattern is striking. However Mladenova's proposed account for this change appears implausible. It is very unclear what it means for a category of nouns, such as mass nouns, to be reinterpreted from divisible to indivisible. Assuming that massnoun-hood is a label for a certain semantic type, does this change mean that there is a type change happening? Generally, it is assumed that different categories of nominals have different but stable mereological types, therefore it might be more appealing to assume that it is actually 'whole' which changes its type over time (from adjective to maximality operator / quantifying determiner), so that, at the end of the process, 'whole' becomes the new 'all'. In any case, as is clear from this short discussion, the empirical pattern literally cries for proper formal semantic analysis.

3. The rise of explicitly formal approaches

3.1 Ontological semantic functionalization: Gergel (2014)

Gergel (2014) discusses the diachronic development of Modern English *most*. *Most* has two uses in Modern English: as a superlative morpheme attaching to a gradable expression (26) and as quantifying (strong) determiner (27):

(26) *The most recent studies supported the criticism.*

(27) *Most studies did not seem to support the criticism.*

There is ongoing debate whether synchronically, quantifying *most* should be analysed as a generalized quantifier in the sense of Barwise & Cooper (1981), establishing relations between sets (e.g. $\text{most}(A,B) = |A \cap B| > |A|/2$); or rather, quantifying *most* should be treated as the superlative of *many* (Hackl 2009), which itself is regarded "as a gradable modifier that modifies plural NPs ranging over pluralities that can be measured in terms of how many atomic parts they are composed of" (Hackl 2009:79). Gergel (2014) remains agnostic with regard to these two analyses, however, he points out that the superlative-quantifier analysis involves a decomposition of the word which is not visible on the surface, and, assuming that this analysis is on the right track, it falls to diachronic semanticists to explain how such a complexity may have evolved over time.

Gergel (2014) analyses the diachronic changes involved in terms of the functionalization of lexical material, proposing a specific and novel type of functionalization. While functionalization often involves changes such as the loss of theta-roles (Roberts 1985) or a type-shift to a higher type (von Stechow 1995), none of this is attested in the case of *most*. Instead, Gergel (2014) proposes a new mechanism called 'ontological semantic functionalization':

(28) MOST [part of [N]] -> MOST [N]

Historically, *most* used to combine with lexical items, specifically, relational nouns with the meaning ‘part’. This noun was first reinterpreted to denote a partition in terms of the semantic ontology involved (e.g. pluralities of individuals). Later, this noun was “incorporated into the meta-language semantically and thus became an integral part of entry [of *most*] after the change”, resulting in the current state where taking recourse to a partition is part of the lexical semantic entry of *most*.

Using corpus evidence, Gergel points out that in Old and Middle English, the precursors of *most* (*maest* and its variants) simply meant ‘of high degree in a scale’, in essence, ‘largest’. Already in Old English and continuing into Middle English, *maest* frequently co-occurred with singular nouns meaning ‘part’ (*dael* and its variants and later the French loanword *part*):

- (29) *scheome is þe meste del as seint Austin seið of ure penitence*
 shame is the largest part as saint Augustine says of our penitence

This usage is preserved in the fossilized expression *for the most part* ‘for the largest part’. A quantificational reading is first attested in Early Modern English (note that on this reading the nouns are plural):

- (30) *Wheras through most Counties of this Realme Horstealine is growen so co~mon*

Gergel (2014) argues that [*maest dael/part* NP₂]-sequences in Old and Middle English were indirect partitive constructions (cf. Corver & van Riemsdijk 2001):

- (31) *and þone mastan dæl þas folces ofslogon*
 and the.ACC largest.ACC part.ACC the.GEN people.GEN killed
 ‘And (they) killed the biggest part/most of the people.’

The Modern English [*most* NP] construction can only be regarded as partitive if one assumes the presence of a silent nominal [*most* Ø_{NP} NP].^{vi} Gergel (2014) regards the stipulation of a silent noun invariably projected with *most* arbitrary; and instead, proposes a structure whereby *most* projects all the relevant semantics (including partitioning) at LF by itself.

Adopting Hackl’s (2009) analysis, at this point Gergel assumes that ‘quantifying’ *most* is not an atomic element, but rather, the superlative of *many*, and its semantics is essentially identical to the semantics of superlatives in general. Namely, the constituent containing the superlative *-est* and a cover comparison class C moves at LF in order to bind a degree variable *d_i*, and the availability of two landing slots accounts for the ambiguity between the relative reading and the proportional reading (cf. also Heim 1985, 1999 and Szabolcsi 1986):

- (32) a. [[John climbed [the [-est C]_i [*d_i* -many mountains]]]
 ‘John climbed more than half of the relevant mountains.’ (proportional reading)
 b. [[John [-est C]_i [climbed [the *d_i* -many mountains]]]
 ‘John climbed more mountains than any other salient individual.’ (relative reading)

In terms of the diachronic change, Gergel (2014) proposes that it was an instance of ontological semantic functionalization, a novel mechanism defined as follows:

- (33) An item undergoes ontological semantic functionalization diachronically iff it changes from compositionally combining with an originally lexical item *I* at a time t_1 towards incorporating a variant of *I* induced in the meta-language of semantic interpretation at a subsequent time t_2 .

In the case of *most*, ontological semantic functionalization involved the transition between (i) compositionally combining with a semi-functional nominal with the meaning ‘part’ to (ii) becoming ontologically enriched so that the partitivity is now hard-wired into the lexical semantics of *most*.

3.2 The Universal Semantic Cycle

3.2.1 Germanic (English, mostly)

An emerging paradigm in the formal study of the diachrony of quantification is due to Beck (2017, 2020). This approach applies the framework of type-theoretical formal semantics to data drawn from electronic historical corpora in a systematic fashion, with the aim of formulating a cross-linguistically valid model of cyclical change in the realm of universal quantification. Works in this paradigm include Beck (2017, 2020, 2022) and Doron (2020). The diachronic study of indefinites in Italian by Degano and Aloni (2021) and the study of FCIs in Hungarian by Halm (2023) is also in part inspired by this paradigm. Bende-Farkas (2014a, 2015) predates and to some extent, anticipates the Universal Semantic Cycle.

The aim of Beck (2017) is to provide a formal semantic account for Haspelmath’s (1995) typological observation that crosslinguistically, free choice relative clauses frequently serve as the diachronic sources of universal quantification. Beck argues that this reinterpretation is in a sense natural since the two sentences below (34ab), while structurally and compositionally different, have roughly the same truth conditions (cf. constant entailments, Beck 2012 and Beck & Gergel 2015, cf. also Eckardt 2009):

- | | | |
|------|---|----------------------|
| (34) | a. <i>I will read [whichever book you recommend]</i> | free choice relative |
| | b. <i>I will read every book (that you recommend)</i> | universal DP |
| | ‘for all x , x a relevant book: I will read x ’ | |

Adopting the generalized quantifier treatment for *every* (Barwise & Cooper 1981, Heim & Kratzer 1998), (34b) can be paraphrased as:

- (35) ‘every individual that has the property of being a book has the property of being read by me’

As far as (34a), the free choice relative clause is concerned, Beck adopts Hirsch’s (2015) analysis. The *wh*-expression *whichever book* introduces Hamblin-alternatives (Hamblin 1973), alternative books, and these alternatives project to the propositional level, yielding alternative propositions of the structure ‘you recommend book x ’. In free choice relatives, these propositions go into the antecedent of a conditional and then the interpretation we receive is:

- (36) ‘Of all alternative propositions p : if p is the proposition that you recommend x (x a book) and p is true, then I will read the book that you recommend’

Beck argues that these two stages (free-choice relative and universal quantifier) are two stages of a four-stage cycle, which she names the Alternative Semantic Cycle (called Universal Semantic Cycle in Beck 2020):

(37) The Alternative Semantic Cycle (also known as the Universal Semantic Cycle)

Stage 1: covert universal quantification over alternative propositions in free-choice relatives:

$$\forall p[p \in \{FCR(x) \mid x \in \text{Alt}(wh)\} \rightarrow \varphi(p)]$$

Stage 2: lexical universal quantification over alternatives:

$$\forall x[x \in \text{Alt}(wh) \rightarrow Q(x)]$$

Stage 3: lexical universal quantification over individuals by universal DP

$$\forall x[P(x) \rightarrow Q(x)]$$

Stage 4: group-denoting DP with possible universal distributive readings

$$\max(\lambda x.P(x)) \in [\lambda z.\forall x[x \leq z \rightarrow Q(x)]]$$

In Stage 1, we have covert universal quantification over alternative propositions triggered by the *wh*-expression. Following Hirsch's (2015) model (inspired by Rawlins's analysis of unconditionals such as *Whichever book you recommend, I will read Harry Potter.*, the meaning of (38) is derived as follows:

(38) *I will read [whichever book you recommend].*

First, the *wh*-expression introduces alternatives:

$$[[whichever\ book]] = \{\text{War and Peace, Harry Potter, Syntactic Structures}\} = \{W, H, S\}$$

These are then projected up, yielding alternative propositions:

$$[[whichever\ book\ you\ recommend]] = \{\text{you recommend W, you recommend M, you recommend S}\}$$

This set of alternative propositions is then placed in the antecedent of a hidden conditional structure (IF), which is then quantified over by a covert universal quantifier over alternative propositions (ALL-ALT_{FCR}), yielding the logical form below (IT stands for a covert pronoun with an E-type semantics: [[IT]] = the book that you recommend):

$$[ALL-ALT_{FCR} [IF [_{FCR} whichever\ book\ you\ recommend]] [I\ read\ IT]]]$$

'Of all alternative propositions *p*: if *p* is the proposition that you recommend *x* (*x* a book) and *p* is true, then I will read the book that you recommend'

In Stage 2, we have lexical quantification over alternatives. Beck's exhibit for this stage is alternative-sensitive universal quantification expressed by the particle *-mo* in Japanese:

(39) *dono gakusei-no okaasan-mo odotta*
 which student-GEN mother-mo danced

'The mother of every student danced.'

for all alternatives x such that $x \in \{y\text{'s mother} \mid y \text{ a student}\}$: x danced

$$\forall x[x \in \text{Alt}(\text{wh}) \rightarrow Q(x)]$$

According to Shimoyama (2001, 2006), *-mo* expresses universal quantification over the alternatives provided by the associate of *-mo*. This associate contains a *wh*-expression, which triggers alternatives that project up to the level of the *mo*-associate.

Stage 3 is the textbook case of lexical universal quantification over individuals by a universal DP:

(40) *The students read every book.*

$$\forall x[P(x) \rightarrow Q(x)]$$

‘every individual entity is such that if it is a book then the students read it.’

Beck argues that Stage 3 is not a stable end-state, pointing out that in Present-Day English, universally quantified phrases are often interpreted non-quantificationally, as group-denoting DPs (cf. Champollion 2010) with possible universal distributive readings. Beck proposes that the bridge that facilitates reinterpretation between Stage 3 and Stage 4 is the quasi-equivalence (i.e., almost identical truth conditions) between (43a) and (43b):

(41) a. *Heidi read every book.*

b. *Heidi read the books.*

In the case of (41a), universal quantification is introduced by the universal quantifier:

(42) *Heidi read every book.*

$$\forall x[\text{book}(x) \rightarrow \text{Heidi read } x]$$

‘For each x such that x is a book, Heidi read x .’

In the case of (41b), the object DP simply denotes a group, and universal quantification is introduced by the distributive operator (DIST) in charge of the distributive interpretation of the predicate (Link 1983):

(43) *Heidi read the books.*

$$\max(\lambda y.\text{book}(y)) \in [\lambda z.\forall x[x \leq z \rightarrow \text{Heidi read } x] =$$

$$\forall x[x \leq \max(\lambda y.\text{book}(y)) \rightarrow \text{Heidi read } x]$$

‘For each x such that x is a part of the largest book group, Heidi read x .’

This means that the truth conditions of (44) and (45) are equivalent, though the ingredients are different. Beck claims that this situation of constant entailments facilitated the reanalysis: a reinterpretation of *everyone* as a group-denoting DP was harmless in that it did not change the truth conditions of such sentences.

Beck argues that the Alternative Semantic Cycle displays a deep parallelism with Jespersen’s (1917) cycle for negation. In both cases, the cycle is related to a certain logical concept (negation, universal quantification). At Stage 1, the lexical expression undergoing the cyclic change does not in itself express this logical concept: rather, it occurs systematically in environments where the concept is expressed “by a complex composition of ingredients”. At Stage 2, expressing the

logical concept becomes a lexical property of the relevant linguistic expression, in conjunction with further ingredients (two negative particles in Jespersen’s cycle, a *wh*-element plus a universal in the AltSemCycle). At stage 3, the semantics of the linguistic expression is in complete overlap with the logical concept (negation, universal quantification). At stage 4, this semantics is weakened – opening up the way for a new cycle or spiral with new linguistic expressions.

The objective of Beck (2020) is to provide an analysis of the quantificational system of Old English in terms of alternative semantics in the vein of Hamblin (1973) and Shimoyama (2006). Old English lacked quantifiers, FCIs and NPIs as separate lexical categories: rather, such interpretations were elicited by so-called bare pronouns such as *hwa* ‘who/what’, *hwelc* ‘which’ or *hwaer* ‘where’, and their prefixed version (the *ge*-paradigm and the *a*-paradigm).

- (44) a. *To hwam ga we*
to whom go we
‘To whom do we go?’ *question*
- b. *(Nellap bi gelyfan) ðeah hwa of deape arise*
though who of death arose
‘they will not believe it, though someone rose from death’ *existential*
- c. *bet ða byssa hwæne hors forlatan*
ordered (he) then (of) warriors whom horse leave
‘he then ordered everyone of the warriors to leave his horse behind’ *universal*
- d. *And a. a. a. to worulde buton æghwīlcum ende Amen*
and ever ever ever to world without A-GE-which end Amen
‘And ever, to time without end.’ (‘...without any end...’) *NPI*
- e. *Sua hwa ðonne sua his lif to biesene bið oðrum, ...*
so who then so his life to example is (to) other
‘Whoever, then, makes his life an example to others, ...’ *FCR*
- f. *Ælc (=a-hwelc) mon mot onsacan frympe & werfabðe,*
a-which man may deny entertainment & slaying
gif he mæg oððe dare.
if he may or dare *FCI*
‘Any man may deny entertainment and slaying, if he is able to or dares.’

Beck (2020) assumes that these pronouns are so-called indeterminate pronouns: pronominal expressions that are used as interrogative pronouns; however, they or their morphological derivations can also be interpreted as indefinites, NPIs, universal or FCIs. The synchronic study of indeterminate (i.e., quantificationally underspecified) pronouns (attested in a variety of languages such as Japanese, Latvian, Hungarian, etc.) goes back to Kuroda (1965) and has been

brought to the fore by Shimoyama (2001) and Kratzer and Shimoyama (2002) (more recent discussions include Mitrovic 2014 and Szabolcsi 2015).

Beck identifies the following possible readings of indeterminate pronouns in Old English: question, free choice relative clause (FCR), free-choice item (FCI), negative polarity item (NPI), universal, existential (see above). The three paradigms (the bare series, the *ge*-series and the *a*-series) differ significantly in terms of the availability of the various readings:

(45)	question	FCR	FCI	NPI	Univ	Exist
bare series	✓	✓	✓		✓	✓
<i>ge</i> -series			✓	✓	✓	✓
<i>a</i> -series			✓	✓	✓	

The outline of Beck's analysis is as follows: indeterminate pronouns are analysed as alternative triggers (in line with earlier proposals such as Hamblin 1973, Shimoyama 2001, Kratzer & Shimoyama 2002). There are also covert operators (question, universal, existential etc.) which interact with the propositional alternatives triggered by the indeterminate pronouns. So sentence interpretation is derived from an interplay between covert propositional operators and alternative-inducing wh-indeterminates. Crucially, there are restrictions as to which pronoun paradigm can associate with which operator – Beck derives these restrictions from the lexical properties of each series.

Inspired by a Hamblin (1973)-style approach to question semantics (where the denotation of a question is a set of propositions, i.e., the set of possible answers), Beck assumes that an indeterminate pronoun introduces lexical alternatives (*who*: {Aelfred, Bede, Caedmon, ...}), which then combine with the syntactic environment (*_ left*) to project propositional alternatives (*who left* {that Aelfred left, that Bede left, that Caedmon left...})

More precisely, adopting the framework of Beck (2006, 2016) a wh-indeterminate has no ordinary semantic value, only an alternative semantic value:

- (46) a. $[[who]]_O = \text{undef (is undefined)}$
 b. $[[who]]_{Alt} = \{x \mid x \in D\} = \{\text{Aelfred, Bede, Caedmon}\}$
 c. $[[who \text{ left}]]_O = \text{undef (is undefined)}$
 d. $[[who \text{ left}]]_{Alt} = \{\lambda w. x \text{ left } w \mid x \in D\} = \{\text{that Aelfred left, that Bede left, that Caedmon left}\}$

The simplest case is the derivation of the interrogative reading. The contribution of the (silent) question operator is to raise the alternative semantic value to the level of ordinary semantic value (as the ordinary semantic value of a question is exactly this set of propositional alternatives); this is necessary as otherwise, we would run afoul of the principle of interpretability (Heim and Kratzer 1998, Beck 2006): *An LF must have an ordinary semantic interpretation*

- (47) a. $[[Q \text{ XP}]]_O = [[XP]]_{Alt}$
 b. $[[Q [who \text{ left}]]]_O = [[who \text{ left}]]_{Alt} = \{\text{that Aelfred left, that Bede left, that Caedmon left}\}$

Alternative-semantics is best-known for its applications to focus (Rooth 1985, 1992); but as Beck notes, it has been used to account for various phenomena such as: free-choice relative clauses

(FCRs, Rawlins 2013, Hirsch 2016, Halm 2023 a.o.), negative-polarity items (NPIs, Krifka 1995, Lahiri 1998, Chierchia 2006 a.o.), free-choice items (FCIs, Menéndez-Benito 2010, Chierchia 2013 a.o.), universal and existential constructions with indeterminate pronouns in Japanese (Shimoyama 2001) and indefinites in German (Kratzer and Shimoyama 2002).

Beck observes that the range of readings of indeterminate pronouns in Old English is, most likely, not coincidental. This motivates her general strategy for accounting for these readings: in each case, the relevant interpretation is elicited via the interaction of an alternative-triggering wh-indeterminate and an alternative-evaluating silent operator:

(48) [Op [... wh-indeterminate ...]]

| _____ |

Interrogatives and FCRs have the same basic structure (in FCRs, there is also universal quantification over the set of propositional alternatives, plus a cover if-operator, see Hirsch 2016 for details):

(49) [Q [... wh-indeterminate ...]]

| _____ |

Universal and existential interpretations are generated when a covert universal (ALL) or existential (EXIST) operator quantifies over the set of propositional alternatives projected by the lexical alternatives introduced by the wh-indeterminate:

(50) a. [ALL /EXIST [... wh-indeterminate ...]]

| _____ |

b. $[[\text{ALL } \text{XP}]]_{\text{O}}(w) = 1$ iff for all $p \in [[\text{XP}]]_{\text{Alt}}$: $p(w) = 1$

c. $[[\text{EXIST } \text{XP}]]_{\text{O}}(w) = 1$ iff for there is a $p \in [[\text{XP}]]_{\text{Alt}}$: $p(w) = 1$

Beck models FCIs as involving generic quantification over possible individuals (cf. Menéndez-Benito (2010)). The lexical alternatives introduced by the wh-indeterminate are projected to the level of propositional alternatives, and these alternatives are evaluated by a cover modal or generic operator. This operator, All-Alt, is modal in nature: it is sensitive to the normal course of the events and it quantifies over possible rather than actual entities introduced by the FCI:

(51) a. [All-Alt [... wh-indeterminate ...]]

| _____ |

b. $[[\text{All-Alt}_R \text{XP}]]_{\text{O}}(w) = 1$ iff for all p such that $p \in [[\text{XP}]]_{\text{Alt}}$:

for all w' such that $R(w')(w)$, $p(w') = 1$, where $R(w')(w)$ iff w' follows the normal course of events in w (R is realistic)

NPIs present an empirical as well as theoretical challenge: since in the bare and the *ge*-series, an existential interpretation is also available, how can one tell apart (weak) NPIs from ordinary narrow scope existentials?

(52) *I didn't eat hmelc cookie.*

'I didn't eat any cookie.'

‘I didn’t eat a cookie.’

$\neg\exists x[\text{cookie}(x) \ \& \ \text{I ate } x]$

Beck observes the following theoretical consideration: if one adopts (i) the alternative semantic analysis of wh-indeterminates advocated for here and (ii) the alternative semantic theory of NPI-licensing (Krifka 1995, Lahiri 1998, Chierchia 2006), then the bare series cannot have a NPI-interpretation. The reason is simple: the licensing mechanism of NPIs according to Chierchia (2006) requires that a wh-indeterminate qua NPI have an ordinary semantic value in addition to an alternative semantic value, and under Beck’s analysis, the bare series does not have an ordinary semantic value.

However, as far as the *ge*-series is concerned, the close analysis of a number of examples leads Beck to conclude that with a reasonable certainty, the *ge*-series did have an NPI-interpretation. Beck argues that this is possible because the contribution of *ge*- is exactly to create an ordinary semantic value for the wh-indeterminate (which in its bare form has no ordinary semantic value):

- (53) a. $[[ge\text{-}XP]]_o = \cup[[XP]]_{Alt}$ (if the elements of $[[XP]]_{Alt}$ are not of type $\langle e \rangle$)
 $= \lambda P.\lambda w.\exists x[x \in [[XP]]_{Alt} \ \& \ P(w)(x)]$ (if the elements of $[[XP]]_{Alt}$ are of type $\langle e \rangle$)
 b. $[[ge\text{-}XP]]_{Alt} = [[XP]]_{Alt}$

The ordinary semantic value of a *ge*-indeterminate is the union of the propositional alternatives elicited and projected by the wh-indeterminate. This explains why the *ge*-series does not have a question or FCR-interpretation (the Q particle needs an operand without an ordinary semantic value), and correctly predicts that the *ge*-series can have a genuine NPI interpretation (as it meets the requirements of EXH). This semantic contribution of the prefix is motivated from the semantic contribution of its likely diachronic source, the free morpheme *ge*, (cf. Beck 242-246 for details, and also Erlewine 2017 for a similar proposal). The NPI-interpretation of the *ge*-series is derived in the standard fashion of Chierchia (2006) (cf. Beck 2020: 248-249 for details and a worked-out example).

The interpretive possibilities of the *a*-series form a proper subset of those of the *ge*-series: FCI, NPI, universal. Beck correlates this to the semantics of the free morpheme *a*, which is the temporal universal quantifier ‘always’:

- (54) $[[always_c]] = \lambda p_{\langle i, st \rangle}.\lambda w. \text{ for all } t \text{ in } C, p(t)(w)=1$

Beck proposes that the prefix *a*- derives from lexical *a*, and it imposes the requirement that the evaluation operator must have universal quantificational force (‘universal agreement’.) Beck models this on negative concord (specifically, Penka (2011)): in each case, there is a (potentially) silent, semantically interpreted element with an interpretable feature (iNeg, iV) and an overt element which does not feed the interpretation (having an uninterpretable feature (uNeg, uV):

- (55) a. $[_{NegP} [_{Neg} Neg [\dots \text{neg-word}[uNeg] \dots]]]$
 $| \text{-----Agree-----} |$
 b. $[ALL/All\text{-}Alt/EXH [\dots a[uV]\text{-ge-hwa} \dots]]$
 $| \text{-----Agree-----} |$

This means that, according to Beck, the diachronic change involved the reinterpretation of lexical *a* (an interpretable exponent of universal quantification, $i\forall$) to prefix *a-* (the exponent of an uninterpretable $u\forall$ feature). This parallels nicely with a development described by Gianollo (2017, 2018): interpretable negator Latin *nec* ($iNeg$) turned into a prefix participating in negative concord with an uninterpretable feature [$uNeg$] in Romance negative indefinites (e.g. *nessuno*). We can also point to another parallel: agreement cycles, which involve the reanalysis of personal pronouns as first clitics and then as agreement-marking morphemes: this change has also been analysed as a transition from interpretable to uninterpretable phi-features (cf. van Gelderen 2023: 140-156).

In terms of diachrony, Beck argues that Old English represented a transition between Stage 1 (covert universal quantification over alternative propositions) and Stage 2 (Universal quantification over individual alternatives) of the Universal Semantic Cycle (see above).^{vii}

Beck (2022) is a follow-up of Beck (2020), charting the diachronic changes in the system of quantification from Old English to Middle English. The Early Middle English (EME) period (ca. 1150-1250) is characterized by a reduction of the combinatory freedom of Old English (OE): while the basic semantics remains the same, the feature system of the *a*-series gets to play a larger role: the operator-trigger relationship becomes more constrained. The *ge*-series falls out of use; the bare series retains some of its quantificational (i.e., non-question) interpretations, but there is a strong decline in frequency and variability: the question interpretation predominates, existential interpretations are still possible (to a more and more limited extent), but universal interpretations are no longer accessible. The *a*-series continues to function with its universal, NPI and FCI readings. Complex forms going back to *aƿre alc* ‘ever each’ start to appear, but remain in the minority as non-complex forms still predominate.

Late Middle English (LME, 1420-1500) is a period of the transition: the flexibility of operator-trigger quantificational system is lost and a new system emerges where there is a direct, one-to-one relationship between the semantic interpretation (question, universal, existential, NPI, FCI) and the morphological form of the *wh*-expression. The non-bare (*a*)-series develops into complex expressions such as *every*, *each* or *every each*. NPI and FCI interpretations of the non-bare series are lost, which means that there arises a one-to-one correspondence between the covert ALL operator and the non-bare series, which sets the stage for members of the non-bare series (such as *every* and *each*) to be reinterpreted as the exponents of a universal operator. Also, in LME, the bare series only has a question interpretation, so there is one-to-one correspondence between the silent operator ($Q[iWH]$) and the alternative trigger (bare-indeterminate $[uWH]$).

In terms of a formal analysis, in EME, non-bare forms carry the uninterpretable feature [$u\forall$] and via universal agreement require an operator of the ALL, All-Alt or EXH type (all of which are $[i\forall]$). The bare series can be evaluated by Q, presumably by EXIST (to a more and more limited extent) but very probably not by ALL.

In LME, the non-bare series continues to have the feature [$u\forall$], however, there is a change in the feature composition of the operators: only ALL retains $[i\forall]$ (i.e., All-Alt and EXH lose their $[i\forall]$ feature): this means that the non-bare series can only be interpreted universally. The bare series acquires an obligatory $[uWH]$ feature and thus, it can only be evaluated by $Q[iWH]$, meaning that only a question interpretation is possible now.

The result of these changes is that in LME, a one-to-one correspondence arises between operator and trigger. This makes a reinterpretation possible: instead of postulating a silent operator rigidly associated with an overt but indirect correlate via agreement; an LME or EModE speaker can

posit a more simple structure, with less silent material: the ex-indeterminate is reinterpreted as the actual exponent of a standard universal quantifier. Consider:

Before reinterpretation (LME):

- (56) a. *uwilc sunne-dei is observed*
 each Sunday is observed
- b. $[[uwilc\ sunne-dei]] = \{x: x\ \text{is a Sunday}\}$ *Alternative trigger*
- c. $[ALL[i\forall] [[uwilc\ sunne-dei][u\forall] \text{ is observed}]]$
- d. For all propositions p : $p \in \{\text{that } x \text{ is observed} \mid x \text{ is a Sunday}\} \rightarrow p \text{ is true}$

After reinterpretation (late LME or EModE)

- (57) a. $[[_{DP}\ ech\ Sunday] \text{ is observed}]$
- b. $[[ech]] = \lambda P.\lambda Q.\forall x[P(x) \rightarrow Q(x)]$ *Generalized quantifier*
- c. $[[ech\ Sunday]] = \lambda Q.\forall x[Sunday(x) \rightarrow Q(x)]$
- d. $\forall x[Sunday(x) \rightarrow observed(x)]$

To conclude, while Old English was a period in transition between Stage 1 and 2 of the Universal Semantic Cycle; Middle English is characterized as a leading up to the transition from Stage 2 to 3. And, arguably, Modern English witnesses a move to Stage 4.

3.2.2 From a collective to a free choice determiner in Biblical Hebrew: Doron 2020

The starting point of Doron's paper is the cross-linguistic observation of Haspelmath (1995) that distributive universal determiners (like English *every*) have two systematic sources (i) free choice determiners and (ii) collective universal determiners like *all*. Focusing on Hebrew *kol*, Doron (2020) proposes the following "Distributivity Cycle" as an account of this pattern:

- (58) a. *kol* Adj/Noun 'entire(ity)'
 —I--->
- b. *kol* collective universal determiner
 —II--->
- c. *kol* free-choice (FC) determiner
 —III--->
- d. *kol* distributive universal determiner

Steps I and II took place in Biblical Hebrew: noun *kol* 'entirety' grammaticalized as the collective determiner *all*, which was then further reinterpreted as a Free Choice/NPI determiner similar to English *any*. Step III happened in Modern Hebrew (or somewhat earlier): free-choice *kol* was reinterpreted as a distributive universal meaning 'every'. Doron argues that the wide range of interpretations *kol* has in Modern Hebrew (all/any/every) is a direct consequence of this process.

Biblical Hebrew (BH) did not have a distributive universal determiner, or indeed, any determiner, being an NP (as opposed to DP) language (cf. Boskovic 2008). As argued by Doron & Meir (2013, 2016), the “article” *han-* was actually not a D but a word-level inflectional prefix marking the emphatic state (as opposed to unmarked absolute state or the construct state marking relationality or possessedness). NPs in argument position underwent type-shift (definite type-shift / iota type-shift or indefinite type-shift / existential closure).

Doron argues that in BH, originally, *kol* was a degree noun, which headed a pseudo-partitive construction and it denoted “the degree of entirety” (i.e., totality).

(59) *kol hā-hākāmīm*

KOL(of) the-skilled.MP

‘all the craftsmen’ (lit. ‘the totality of the craftsmen’) (Ex. 36:4)

The original noun-hood of *kol* is supported by its distribution and indeed, traditional grammars of the Bible also classified *kol* as an ‘abstract substantive denoting totality’ (Joüon 1923: §139e). That is, *kol* NP at this stage was non-quantificational (and non-distributive): this is evidenced by (i) the lack of scope ambiguity, (ii) the fact that the cardinality of *kol* NP could be predicated about and (iii) agreement patterns among other things). *Kol* NP was strictly speaking not interpreted as ‘all N’ but rather as ‘the totality of Ns’: At this stage, *kol* NP denoted the entirety/totality of a group or indeed of an individual or a mass term. More formally, it was a non-quantificational operator that applied to “a NP which denote[d] a substance, mass or count, singular or plural, and yield[ed] a portion of the NP substance that consist[ed] of the entirety of those individuals whose parts satisfy[ed] NP. Hence *kol* map[ped] a predicate P to the set of individuals, atoms or sums, satisfying *P.” (Doron 2020:11)

(60) $[[kol]] = \lambda P.\lambda x.*P(x)$

where *P denotes the minimal divisive predicate of Krifka (1989) which includes P: P itself where P is divisive per se, or the pluralized version of P

In argument position, the default interpretation of *kol* NP was definite: the maximal individual $\iota x.kol[[NP]](x)$ satisfying it. In downward entailing (DE) environments, however, the indefinite interpretation (arising via via predicate modification and existential closure) was stronger and thus favoured. This was due to the stronger meaning hypothesis (*Pick the stronger meaning*, Dalrymple et al. 1994). Since this indefinite interpretation of *kol* NP only arose in DE environments, indefinite type-shifted *kol* NP in BH could be regarded as a negative polarity item (NPI).

Since *kol* was not distributive, one might wonder what its exact contribution was: in other words, what was the difference between NP and *kol* NP? Doron argued that *kol* was a slack regulator (cf. Brisson 1997, Krifka 2006, Lasersohn 1999, Schwarz 2013), it reduced (eliminated) exception-tolerance:

(61) *qārōaʕ ʔeqraʕ ʔet ham-mamlākā mē-šālekā ... raq ʔet kol*
 tear.INF tear.MOD.1S ACC the-kingdom from-over.2MS ... but ACC KOL
ham-mamlākā lō ʔeqraʕ – šebet ʔebad ʔeten
 the-kingdom NEG tear.MOD.1S – tribe.MS one.MS give.MOD.1S
li-bn-ekā
 to-son-POSS.2MS

‘I will surely tear the kingdom away from you... However, I will not tear away the whole kingdom; I will give one tribe to your son.’ (1Kings 11:13)

Ham-mamlākā ‘the kingdom’ involves the pragmatic looseness characterised as slack: depending on the context, *I will tear the kingdom away*. is taken to be true even if some part of the kingdom is not actually taken away. *Kol ham-mamlākā* strictly means the totality of the kingdom, every single tribe of it.

The environments in which indefinite-*kol*-NP can be found in BH include downward entailing environments that typically license NPIs (negation, questions, complements of adversative verbs, before PPs, comparative PPs, antecedents of conditionals) and also environments that typically license FCIs *strictu sensu* (generics, free relatives). Further on, Doron shows that in the scope of existential modals, *kol* NP had a free choice reading. *Kol* NP was also found in the scope of imperative/commissive modal operators, as well as being modified by a relative clause: in such cases, it was interpreted like *whatever/whoever*, a free relative with FC interpretation.

Doron argues that as far as BH was concerned, the FC reading of indefinite-*kol*-NP arose via the mechanism described by Kratzer & Shimoyama (2002): the distribution requirement qua conversational implicature. Later on, argues Doron, this FC interpretation was semanticized and reanalysed as being part of the lexical meaning of *kol* – in line with Eckardt’s (2006: 236) conception of grammaticalization as the semanticization of pragmatically inferred information, leading to the reanalysis of *kol* as a free-choice item (‘any’).

Finally, Doron turns to Step III of the distributivity cycle, which leads to the current situation in Modern Hebrew (the change actually may have occurred in Rabbinical Hebrew already), where *kol* has a distributive universal reading in addition to the free choice item reading:

(62) *kol* NP = any NP_{et} / every (distributive universal) NP_{et}

Doron does not dwell much on the specifics of this change, other than stating that Step III must have happened in line with Beck’s (2017) proposal: “the conjunction of the alternative propositions which underlies FC readings developed into universal quantification over individuals”.

Another important change is that in post-Biblical Hebrew, *ba-* was reanalyzed as a definite determiner of category D – so definite noun phrases are no longer (iota-shifted) NPs but true DPs (Doron & Meir 2016). This means that in Modern Hebrew, there is a difference in interpretation of *ba*-prefixed and *ba*-less Ns with *kol*, with plurality also playing a role. Consider:

- (63) a. *ba-banbala lo hithayba le-qabel et kol ba-tlunot*
the-administration NEG committed to-accept ACC KOL the-complaints
‘The administration did not commit to accept all/*any complaints.’
- b. *lo nimce’u kol tlunot mucdaqot*
NEG found KOL complaint.FP justified.FP
‘There weren’t any justified complaints found.’
- c. *ba-banbala lo hithayba le-qabel et kol tluna*
the-administration NEG committed to-accept ACC KOL complaint
‘The administration did not commit to accept every/any complaint.’

To summarize:

- (64) a. *kol*/DP = all DP_e partitive structure expressing totality (quasi-collective universal) (63a)
- b. *kol*/NP_{plural} = any NP_{et} free-choice item (“remnant of Biblical syntax”) (63b)
- c. *kol*/NP_{singular} = every/any NP_{et} distributive universal / free-choice item (63c)

This means that in Modern Hebrew, all the stages of the cycle are preserved and visible.

3.3 From indeterminates to A-quantification to D-quantification in Hungarian: Bende-Farkas (2014a, 2015, 2020)

In a series of studies, Bende-Farkas (2014a, 2015, 2020) provides a corpus-driven empirical study of universal quantification in Old and Middle Hungarian. Three main themes emerge: (i) the transition from Stage 1 to Stage 3 of the Universal Semantic Cycle, (ii) the emergence of D-quantification and (iii) an ongoing ‘whole’-‘all’-‘every’ cycle.

3.3.1 From Stage 1 to Stage 3 of the Universal Semantic Cycle

Bende-Farkas (2014a, 2015) argues that Early Old Hungarian, the period which directly predates our earliest written sources, in all probability had a system of bare indeterminate pronouns being bound by silent propositional operators (much like the Old English system described by Beck 2017, 2020, see above). Remnants of this system are sporadically detectable in Late Old Hungarian texts: bare pronouns can be interpreted as NPIs in the scope of a negator, as existential/FCIs in the antecedent of a conditional or as an existential bound by existential closure:

- (65) a. *aʒ egjebekrewl nem tudok mjtt*
the other.PL.from not know.1SG what.ACC
‘I do not know anything about the rest.’
- b. *Ha ky kerdenee honnan volt aʒ [...]*
if who ask.COND.3SG whence was it
‘If someone/anyone were to ask why it was the case that [...]’
- c. *ky kezeyt, ky edes zemeyt [...] apolgyaak vala [...]*
who hand.POSS.3SG.PL.ACC who sweet eye.POSS.3SG.PL.ACC kiss.PAST.3PL PAST
‘Some were kissing his hands, some were kissing his eyes.’

Bare pronouns also featured in so-called correlatives^{viii} (correlatives resemble free relatives in their syntax and semantics, the major difference being that (i) correlatives can include several relative expressions and (ii) each relative expression has a so-called correlate, typically, a demonstrative in the matrix clause). Such usage is preserved well into Modern Hungarian in proverbs such as:

- (66) *ki mint veti ágyát, (aʒ) úgy alussza álmát*
who how set.3SG bed.3G.ACC that so sleep.3SG dream.3SG.ACC

‘The way someone sets their bed determines how they sleep.’ (Lit. ‘Whoever sets his bed in whatever way, that is how (s)he will sleep.’)

Since correlatives are close to free relatives, one could plausibly treat correlatives in the way proposed by Beck (2020) wrt to free relatives in English, with the propositional alternatives triggered by the *wh*-indeterminate going into the antecedent of a covert conditional structure, possibly involving the presence of a silent *Q*-operator as well. Belyaev & Haug (2020) have proposed that cross-linguistically, correlatives often derive from paratactic constructions. As Belyaev & Haug argue, such a scenario is eminently plausible wrt Hungarian:

- (67) *ki mint veti ágyát, (az) úgy alussza álmát*
 who how set.3SG bed.3G.ACC that so sleep.3SG dream.3SG.ACC
- i. ‘Someone sets their bed somehow. This (same) person sleeps that way (in a corresponding way). *paratactic reading*
- ii. ‘Whoever sets his bed in whatever way, that is how (s)he will sleep.’ *correlative reading*

Late Old Hungarian saw the emergence and solidification of particle plus indeterminate combinations:

(68) Late Old Hungarian particle+*wh* indeterminate series

	universal	existential	correlative	FCI	NPI
<i>vala</i> + <i>wh</i> -indeterminate:		✓	✓	✓	
<i>se(m)</i> + <i>wh</i> -indeterminate					✓
<i>akár</i> + <i>wh</i> -indeterminate			✓	✓	
<i>minden</i> + <i>wh</i> -indeterminate	✓				

This gradual process led to the current situation in Modern Hungarian, where bare pronouns can only be interpreted as questions words. As far as particle+*wh*-indeterminate combinations are concerned, Bende-Farkas takes these particles to be the exponents of uninterpretable features which have to be checked by a higher covert operator, with the exception of *minden*-, which is an exponent of a universality/maximality operator (see the discussion of Bende-Farkas 2014 below). This change is thus very similar to the reduction of the combinatorial possibilities in the Late Middle English period discussed by Beck (2022).

As far as the origins of these particles are concerned, Bende-Farkas (2014a) presents a detailed analysis of the emergence of *minden*- (see discussion below). Concerning the *se(m)*-series, É. Kiss (2014) proposes that this is derived from a combination of *es* ‘focus particle’, *nem* ‘negator’ and the indeterminate pronoun (adjacent with the former in the preverbal focus position):

- (69) [*es* ‘focus particle’ + *nem* ‘negator’] + *ki* ‘who’ -> *senki* ‘who[uNeg]’

In later work, Halm (2023) offers a formal diachronic analysis for the emergence of the two FCI-paradigms: the *akár*-paradigm in Old Hungarian (first FCR, then FCI: *akárki* ‘whoever -> anyone’) and the *bár*-paradigm in Middle Hungarian (first FCR, then FCI: *bárki* ‘whoever -> anyone’). The gist of Halm’s proposal is that in Late Old Hungarian, *akár* was the exponent of an *if*-operator in unconditionals and thus could scope over one or more *wh*-indeterminates, supplying the proposition set involved in unconditionals (cf. Rawlins 2013):

- (70) *akár mit mondanak, én megyek a magam útján*
 IF.UNCOND what.ACC say.3PL, I go.1SG the own.1SG way.3SG.ACC

‘If they say A, I will go my own way, if they say B, I will go my own way...’

In Middle Hungarian, adjacent *akár* plus wh-indeterminate combinations were reinterpreted as lexemes denoting properties, and their sentence initial position as the syntactic focus position (the canonical position of question words). As proposed by Hirsch (2016) in the context of English, the covert Q-morpheme turns the property headed by the *akár*-expression [$\lambda x. \lambda w. \text{they say } x \text{ in } w$] into a set of propositions, yielding the free relative interpretation:

- (71) *akármit mondanak, én megyek a magam útján*
 whatever.ACC say.3PL, I go.1SG the own.1SG way.3SG.ACC

‘Whatever they say, I will go my own way.’

Later, *akár*-expressions were further reinterpreted from free choice relatives (‘whoever’) to FCIs (‘anyone’).

Since Bende-Farkas stops at the end of the Late Old Hungarian period, mapping the changes from then on to current Modern Hungarian remains an open challenge. Most synchronic analyses of Modern Hungarian assume (either implicitly or explicitly) that particle+pronoun combinations such as *minden-ki* ‘everyone’, *vala-ki* ‘someone’, *bár-ki* ‘anyone’, *sen-ki* ‘no-one’ do not involve quantification by covert operators over propositional alternatives, and also, that these lexemes are no longer compositional (though see Szabolcsi 2015 for a different view):

- (72) a. *minden-ki* ‘everyone’ universal quantifier (Barwise & Cooper)
 b. *vala-ki* ‘someone’ positive polarity (Heimian) indefinite
 c. *bár-ki* ‘anyone’ free-choice-item (intensional indefinite)
 d. *sen-ki* ‘no one’ negative universal quantifier

If this is indeed the case, Hungarian can be said to have undergone a change from Stage 1-2 to Stage 3 in the Universal Semantic Cycle.

3.3.2 From A-quantification (*mind* ‘all’) to D-quantification (*minden* ‘every’)

Bende-Farkas (2014a) discusses the diachrony of two quantifiers expressing maximality or universality: *mind* ‘all’ (an A-quantifier which resembles English *all* or French *tous/toutes*) and *minden* ‘every/everything/everyone’ (a D-quantifier akin to English *every*).

Bende-Farkas argues that prior to the emergence of *minden* ‘every’, Late Old Hungarian did not have D-quantification. Universal and universal-like readings were elicited by (i) means of the A-quantifier *mind* (cf. 69 below), by (ii) other A-quantifiers such as *-nkéd*-suffixed numerals and nominals (*egy-enkéd* one-PLACT ‘one by one’, *fej-enkéd* head-PLACT ‘one by one’, *nap-onkéd* day-PLACT ‘day by day’^{ix}) and (iii) covert quantification over propositional alternatives projected by wh-indeterminates (see 3.3.1 above).

Drawing on corpus evidence, Bende-Farkas characterizes *mind* ‘floating all’^x in Old Hungarian as an A-quantifier:

- (73) *Es vimagguc mend szentucut.*

and worship.1PL all saint.PL.ACC

‘and let us pray to all the saints’ Funeral Sermon and Prayer (1192-1995)

Syntactically, Bende-Farkas analyzes *mind* ‘floating all’ as an adverbial adjoined to the DP or the (extended) VP. Semantically, Bende-Farkas argues that *mind* ‘floating all’ was not a proper quantifier with Restrictor and Nuclear Scope, rather, it was a maximality operator: a predicate modifier which says that the predicate P holds of the maximal individual $U(X)$ constructed from the collection X (X being contributed by the DP-associate of *mind*) – so essentially, it is a definite description:

(74) *mind* (non-distributive): $\lambda P.\lambda X.[P(U(X))]$

Though the notation is different, this is identical to the ‘every’ part of group-denoting *everyone* discussed by Beck (2017, 2020) in Stage 4 of the Universal Semantic Cycle (see above).

Bende-Farkas also sketches a compositional analysis of *mind*. *Mi* ‘what’ is taken to be a wh-indeterminate triggering individual lexical alternatives, and *-nd* is assumed to denote “a functor which takes these alternatives as input and returns the entire set (or the greatest individual ‘built’ from the members of that set)”. Since, as we have seen above, Early Old Hungarian had a system similar to Stage 1 of the Universal Semantic Cycle of Beck (2017, 2020) (remnants of which survived well into Late Old Hungarian), a different potential analysis also appears feasible, whereby *-nd* could be analysed as an overt exponent of Beck’s ALL-operator, or maybe an exponent of $[u\forall]$. (This would arguably make *mind* a representative of Stage 2 of the USC.)

Bende-Farkas argues that the *mind* ‘all’ $\rightarrow$ *minden* ‘every’ change (which took place in Late Old Hungarian) was an instance of “type-driven change”: “a diachronic application of the [synchronically posited] type-shifting principles of formal semantics” (cf. Partee (1987), Partee-Rooth (1983), Winter (2001); see also Chapter 7 in Eckardt (2006)). The reinterpretation happened in four stages:

(75) From *mind* ‘floating all’ to *minden* ‘every(thing/one)’

Stage 1: <i>mind</i> ‘floating all’	VP-modifier	$\langle\langle e,t\rangle,\langle e,t\rangle\rangle$
Stage 2: <i>minden</i> _{Adj/Adv} ‘entire(ly), full(y)’	adverb/adjective	$\langle\langle e,t\rangle,\langle e,t\rangle\rangle$
Stage 3: <i>minden</i> _D ‘every’	quantifying determiner	$\langle\langle e,t\rangle,\langle\langle e,t\rangle,t\rangle\rangle$
Stage 4: <i>minden</i> _{DP} ‘everything/everyone’	quantifying DP of type	$\langle\langle e,t\rangle,t\rangle$

Stage 2 thus crucially mediates between Stage 1 (‘all’) and stage 3 (‘every’). *Minden* ‘entire(ly), full(y)’ (ambiguous between adjective and adverb) is argued to be derived from *mind* ‘floating all’ via the suffix *-en*. There is some limited empirical evidence suggestive of an adjectival/adverbial *minden*:

(76) *Ez zamos zent napokban myndden eletwnket meg jobbobok*
 this numerous holy days-in entire life.POSS.1PL.ACC PRT improve.1PL

‘During these many feast days we should improve our entire life.’

In (76), a quantifier reading of *myndden* is highly implausible given that each human has only one life (in the relevant sense). However, it remains unclear how exactly a combination of the floating quantifier *mind* with a manner/cardinality suffix *-en* (see discussion below) would derive the

required adjectival reading (no synchronic or diachronic compositional analysis is given). This cline is also rather different from the more generally attested pattern:

- (77) a. whole → all → every (English, German, Greek etc., Haspelmath 1995 a.o.)
 b. all → whole → every (Bende-Farkas 2014)

The final step in the process was the reinterpretation of *minden*_D as *minden*_{DP} by saturating the first argument of *minden*_D with the predicate ‘thing/individual’:

- (78) *méguon menden varost és medent a földön*
 PRT.take.PST.3SG every town.ACC and everyone.ACC the land.on
lakozot
 dwell.PART.ACC
 ‘He conquered every town and everyone living in the region.’

This last step is atypical in that it goes against the Head-Preference Principle (van Gelderen 2004) and the general tendency in diachronic change for higher types (von Fintel 1995).

Bende-Farkas also carries out a detailed study of the syntactic side of the change: while *mind* ‘floating all’ was adjoined to the DP or the (extended) VP, quantifying determiner *minden* is integrated into the DP. In the same volume, É. Kiss (2014b) analyses the external syntax of quantifiers in Late Old Hungarian, showing that (similarly to Modern Hungarian) they occupied dedicated scope-taking positions on the left periphery of the sentences, having undergone quantifier-raising (in the case below, we witness over Q-raising from an embedded non-finite clause into the functional left periphery of the matrix clause):

- (79) *Es [mend paradisumben uolov gimilcictul]_i munda neki*
 and all Paradise-in be.PART fruits.from tell.PST.3SG DAT.3SG
elnie t_i
 live.INF.3SG
 ‘And he told him to eat all fruits in Paradise’

The question of distributivity is also examined in much empirical detail. In Modern Hungarian (cf. Szabolcsi 1997), *minden* ‘every’ is inherently distributive. However, Bende-Farkas argues that in Old Hungarian, *minden* ‘every’ was non-distributive, representing a bridge between Old Hungarian *mind* ‘floating all’ and Modern Hungarian *minden* ‘every (distributive)’ in terms of its semantics:

- (80) *minden* (non-distributive) $\lambda P.\lambda Q.[Q(UP)]$ Old Hungarian

This formula is taken to mean that the predicate Q is true of the maximal individual with property P. Bende-Farkas argues that here, we already have a tripartite structure of sorts but we have not yet gone all the way to classical Barwise & Cooper (1981)-style generalized quantifier semantics, something which is reached only later:

- (81) *minden* (distributive): $\lambda P.\lambda Q.\forall x.[P(x) \rightarrow Q(x)]$ Modern Hungarian

This cline is elegant and makes sense theoretically, and the empirical evidence, while somewhat sparse, is also compatible with this account. Note that the transition from *mind* ‘maximality operator’ to *minden* ‘distributive every’ is the exact opposite of what is presumably happening in Modern English (Stage 3 to 4 of the Universal Semantic Cycle) – presenting a challenge to the unidirectionality (cf. WBCDL026) and/or the universality (cf. WBCDL016) of the Universal Semantic Cycle.

3.3.3 ‘Whole’ -> ‘all’ -> ‘every’ in Transylvanian and Csángó Hungarian

Bende-Farkas (2020) analyses an ongoing ‘whole’->‘all’->‘every’ type reinterpretation process in Transylvanian and Csángó (or Moldavian) Hungarian. The starting observation is that while *egész* ‘whole’ is used in every regional dialect of Modern Hungarian as an adjective meaning ‘whole’; in Moldavian (Csángó) Hungarian and in colloquial Transylvanian Hungarian, the expression *az egész-en* (lit. the whole-PRED) is used as an ‘all’-type floating quantifier, associating with plural individuals (count nouns, pronouns):

(82) *Megérkezünk az egész-en.*

PRT.arrive.PAST.1PL the whole-PRED

‘We have all arrived.’ (my glosses, PRED stands for predicativizing suffix, see discussion below)

In addition to this, in these variants of Hungarian, *az egész* (lit. the whole) can also be used as a collective as well as a distributive universal determiner associating with count nouns:

(83) a. *Egy fán volt az egész madár.*

One tree.on was the whole bird

‘All the birds were in one tree.’

b. *Az egész politikusnak van autója.*

the whole politician.DAT is car.3SG

‘Every politician has a car.’

Bende-Farkas’s account for this synchronic pattern is comprised of two elements: (i) the diachronic process whereby *egész* ‘whole’ was reanalysed as *az egész* ‘all/every’ and (ii) the synchronic, compositional mechanism by which the A-quantifier *az egész-en* ‘floating all’ is derived from the D-quantifier *az egész* ‘all’. The diachronic process is reconstructed using a varied range of resources (the Old Hungarian Corpus (Simon & Sass 2012, Simon 2014), the Historical Dictionary of Transylvanian (Middle to Early Modern Hungarian) as well as hand-collected data from the internet reflecting current colloquial usage. Based on this analysis, the following stages are identified:

(84) Stages of the ‘whole’->‘all’ reinterpretation in Transylvanian and Csángó Hungarian

Stage 1: with abstract nouns (often denoting institutions) and nouns denoting collective entities, *egész* ‘whole’ gets reinterpreted as *egész* ‘entire, all (collective)’:

(a) *egész emberi nemzet*

whole human nation

‘the whole humanity’ -> ‘all humanity’

Stage 2: these expressions are reinterpreted pointwise, due to metonymy

(b) *egész Jeruzsálem*

whole Jerusalem

‘all Jerusalem’ -> ‘everyone in Jerusalem’

Stage 3: the ‘all’-construal is extended to count nouns

(c) *až egész város* --> *až egész házak*

the whole city the whole houses

‘all the city’ ‘all the houses’

Stage 1 held in Old Hungarian, Stage 2 was reached from Late Old to Early Middle Hungarian. These stages affected all regional dialects of Hungarian. Stage 3 was only reached in Transylvania and the Csángó dialect region. Bende-Farkas (2020:63-64) also claims that *až egész* has been further reinterpreted from D to DP for some innovative speakers, however, the evidence provided is inconclusive.

Synchronically speaking, Bende-Farkas relates the obligatory presence of the definite article in *až egész(en)* to Matthewson’s (2001) claim that strong universal determiners always involve a (covert or overt) D head. Following Csirmaz & Szabolcsi (2012) and (Schvarcz 2022), Bende-Farkas assumes that the *-en* suffix has the function of turning a quantifier into a predicative element:

(85) a. *A legtöbb diák szorgalmas volt.*

the most student diligent was

‘Most students were diligent.’

b. *A szorgalmas diákok voltak a legtöbb-en.*

the diligent students were the most-PRED.

‘The diligent students were the most numerous.’

Bende-Farkas thus argues that *až egészen* ‘floating all’ is derived from *až egész* ‘every’ in the same way as other adverbial quantifiers such as *a legtöbb* are derived from D-quantifiers such as *a legtöbb*. That is, this process is synchronic and compositional.

4. Conclusion

This chapter surveyed the progress of the formal diachronic study of quantification in the last couple of decades. Important advances have been made in terms of identifying cross-linguistically valid patterns of change such as the whole-all cycle, the Universal Semantic Cycle, the transition from D-quantification to A-quantification or the reinterpretation of pluractionals as universal quantifiers. There are, however, significant challenges ahead. This includes the broadening of empirical coverage (by extending the discussion to a more diverse and broader set of languages) and the deepening of our understanding of the changes identified (by a more thorough application of the toolkit of formal semantics and also by placing a stronger focus on syntax-

semantics interface phenomena and the interplay of diachronic change in the realms of syntax and semantics).

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References

- Alonso-Ovalle, Luis, and Paula Menéndez-Benito. 2020. “Free choice and modal indefinites.” In *The Wiley Blackwell Companion to Semantics*, edited by Lisa Matthewson, Cecile Meier, Hotze Rullmann, and Thomas Ede Zimmermann. Wiley-Blackwell.
- Barwise, John, and Robin Cooper. 1981. “Generalized quantifiers and natural language.” *Linguistics and Philosophy* 4: 159–219.
- Beck, Sigrid. 2012. “Pluractional comparisons.” *Linguistics and Philosophy* 35, 57–110.
- Beck, Sigrid. 2017. “An alternative semantic cycle for universal quantifiers.” *Wellington Working Papers in Linguistics* 23: 5–13.
- Beck, Sigrid. 2020. “Indeterminate Pronouns in Old English.” *Journal of Comparative Germanic Linguistics*. URL: https://archive.nytd.hu/fods7/slides/Beck_talk.pdf
- Beck, Sigrid. 2021. “Multiple Events and ‘N Preposition N’.” In *The Oxford Handbook of Grammatical Number*, edited by Patricia Cabredo Hofherr and Jenny Doetjes. Oxford University Press.
- Beck, Sigrid. 2022. “From Old English to Middle English: Universal Quantifiers.” Talk given at FoDS7 (*Formal Diachronic Semantics* 7), Research Institute for Linguistics, Budapest.
- Beck, Sigrid, and Remus Gergel. 2015. “The diachronic semantics of English *again*.” *Natural Language Semantics* 23: 157–203.
- Belyaev, Oleg, and Dag Haug. 2020. “The genesis and typology of correlatives.” *Language* 96: 874–907.
- Bende-Farkas, Ágnes. 2014a. “From A-quantification to D-quantification: Universal quantifiers in the sentence and in the noun phrase.” In *The evolution of functional left peripheries in Hungarian syntax*, edited by Katalin É. Kiss. Oxford: Oxford University Press. 83–121.
- Bende-Farkas, Ágnes. 2014b. “Minden valaki — Az ómagyar kori vala-kifejezések értelmezéséhez [Every someone — On the interpretation of vala-expressions in Old Hungarian].” In *Nyelvelmélet és diakronia 2 [Language theory and diachrony 2]*, edited by Katalin É. Kiss and Attila Hegedűs. Pécs: PPKE BTK. 10–41.

- Bende-Farkas, Ágnes. 2015. "The landscape of universal quantification in Old Hungarian." *Acta Linguistica Hungarica* 62: 223-261.
- Bende-Farkas, Ágnes. 2020. "'The Whole of Us Were There': A Little-known Grammaticalisation Process in Hungarian." *Finno-Ugric Languages and Linguistics* 8: 27-75.
- Bende-Farkas, Ágnes. 2024. "Quantification." Ms, Hungarian Research Centre for Linguistics.
- Berman, Steve. 1990. "Towards the Semantics of Open Sentences: WhPhrases and Indefinites." In *Proceedings of the Seventh Amsterdam Colloquium*, edited by Martin Stokhof and Leen Torenvliet. University of Amsterdam: ITLI. 53-78.
- Bošković, Željko. 2008. "What will you have, DP or NP?" *Proceedings of NELS* 37. 101–114.
- von Bremen, Klaus. 1983. "Question-words: a study in the syntax of relativization, free relatives, pseudo-cleft sentences and certain indefinite pronouns." PhD dissertation. Stockholm University.
- Brisson, Christine. 1997. "On definite plural NP's and the meaning of 'all'". *Proceedings of SALT VI*. 55–72.
- Champollion, Lucas . 2010. "Parts of a whole: Distributivity as a bridge between aspect and measurement." PhD dissertation. University of Pennsylvania.
- Chesterman, Andrew. 1991. "On Definiteness: A Study with Special Reference to English and Finnish." Cambridge: Cambridge University Press.
- Chierchia, Gennaro. 2006. "Broaden your views: Implicatures of domain widening and the "logicality" of language." *Linguistic Inquiry* 37: 535–590.
- Chierchia, Gennaro. 2013. "Logic in grammar." Oxford: Oxford University Press.
- Corver, Norbert, and Henk van Riemsdijk. 2001. "Semi-lexical categories." In *Semi-lexical categories. The function of content words and the content of function words*, edited by Norbert Corver and Henk van Riemsdijk. Berlin & New York: Mouton de Gruyter. 1-22.
- Csirmaz, Anikó, and Anna Szabolcsi. 2012. "Quantification in Hungarian." In *Handbook of quantifiers in natural language*, edited by Edward L. Keenan and Denis Paperno. Dordrecht: Springer. 399–467.
- Dalrymple, Mary, Makoto Kanazawa, Sam Mchombo, and Stanley Peters. 1994. "What do reciprocals mean?" *Proceedings of SALT IV*, 61–78.
- Dayal, Veneeta. 1998. "Any as inherently modal." *Linguistics and Philosophy* 21: 433–476.
- Degano, Marco, and Maria Aloni. 2022. "Indefinites and free choice: When the past matters." *Natural Language & Linguistic Theory* 40: 447-484.
- Doron, Edit. 2020. "From a collective to a free choice determiner in Biblical Hebrew." In *Quantification and scales in change*, edited by Remus Gergel and Jonathan Watkins. Berlin: Language Science Press. 1–31.
- Doron, Edit, and Irit Meir. 2013. "Amount definites." *Recherches Linguistiques de Vincennes* 42: 139–165.
- Doron, Edit, and Irit Meir. 2016. "The impact of contact languages on the degrammaticalization of the Hebrew definite article." In *Language contact and the development of Modern Hebrew*, edited by Edit Doron. Leiden: Brill. 281–297

- Dowty, David R. 1987. "Collective predicates, distributive predicates, and *all*." In *Proceedings of ESCOL 4*, edited by Ann Miller and Joyce Powers. Columbus, OH: Ohio State University. 97–115.
- Eckardt, Regine. 2006. "Meaning change in grammaticalization: An enquiry into semantic reanalysis." Oxford: Oxford University Press.
- Erlewine, Michael Yoshitaka. 2017. "Quantifying over alternatives with Toba Batak *manang*". Talk given at AFLA 24.
- Erlewine, Michael Yoshitaka, and Hadas Kotek. 2016. "Even-NPIs in Dharamsala Tibetan." *Linguistic Analysis: Special Issue on South Asian Morphosyntax* 40(3–4): 129–166.
- É. Kiss, Katalin (ed.). 2014a. "The evolution of functional left peripheries in Hungarian syntax." Oxford: Oxford University Press.
- É. Kiss, Katalin. 2014b. "The evolution of functional left peripheries in the Hungarian sentence." In *The evolution of functional left peripheries in Hungarian syntax*, edited by Katalin É. Kiss. Oxford: Oxford University Press. 9–55.
- von Fintel, Kai. 1994. "Restrictions on quantifier domains." PhD dissertation, University of Massachusetts.
- von Fintel, Kai. 1995. "The formal semantics of grammaticalization." *Proceedings of NELS 25*, 175–189.
- Frisk, Hjalmar. 1970. "Griechisches etymologisches Wörterbuch." Bd. 1-2. Heidelberg: Carl Winter.
- van Gelderen, Elly. 2004. "Economy, Innovation, and Prescriptivism: From Spec to Head and Head to Head." *Journal of Comparative Germanic Linguistics* 7: 59–98.
- van Gelderen, Elly. 2011. "The linguistic cycle: Language change and the language faculty." Oxford: Oxford University Press.
- van Gelderen, Elly. 2024. "The linguistic cycle: Economy and renewal in historical linguistics." Routledge.
- Gergel, Remus. 2014. "Most historically." In *Language Change at the Syntax-Semantics Interface*, edited by Chiara Gianollo, Agnes Jäger, and Doris Penka. Mouton De Gruyter. 101-124.
- Gianollo, Chiara. 2017. "Grammaticalization parameters and the retrieval of alternatives: Latin *nec* from discourse connector to uninterpretable feature." Talk given at FoDS2 (Formal Diachronic Semantics 2), Universität des Saarlandes.
- Gianollo, Chiara. 2018. "Indefinites between Latin and romance." Oxford: Oxford University Press.
- Hackl, Martin. 2009. "On the grammar and processing of proportional quantifiers: 'most' versus 'more than half'." *Natural Language Semantics* 17: 63–98.
- Halm, Tamás. 2016a. "The Syntactic Position and Quantificational Force of FCIs in Hungarian." *Acta Linguistica Academica* 63: 241-276.
- Halm, Tamás. 2016b. "The Grammar of Free-Choice Items in Hungarian." PhD Dissertation, Pázmány Péter Catholic University.
- Halm, Tamás. 2023. "Az akár- és a bár- névmási paradigmák diakrón szemantikája. [Want, unconditionals and scalar particles: the sources of free-choice items in Hungarian.]" In *Nyelvelmélet és Diakronia 5 [Linguistic Theory and Diachrony 5]*, edited by Katalin Balogné Bérces, Attila László Nemesi, and Balázs Surányi. Pázmány Péter Catholic University: Budapest.
- Halm, Tamás, and Ágnes Bende-Farkas. 2024. "Pluractionality and diachrony: the case of *-nkéd* in Old and Middle Hungarian." *Acta Linguistica Academica*.

- Hamblin, Charles. 1973. "Questions in Montague English." *Foundations of Language* 10: 41–53.
- Haspelmath, Martin. 1995. "Diachronic Sources of All and Every." In *Quantification in Natural Languages*, edited by Emmon Bach, Eloise Jelinek, Angelika Kratzer, and Barbara H. Partee. Dordrecht: Kluwer Academic Publishers. 363–82
- Haspelmath, Martin . 1997. "Indefinite pronouns." Oxford: Clarendon Press.
- Heim, Irene. 1985. "Notes on comparatives and related matters." Ms., University of Texas at Austin.
- Heim, Irene. 1990. "E-Type Pronouns and Donkey Anaphora." *Linguistics & Philosophy* 13: 137-177.
- Heim, Irene. 1999. "Notes on superlatives." Ms., Massachusetts Institute of Technology.
- Heim, Irene and Kratzer, Angelika. 1998. "Semantics in generative grammar." Oxford: Blackwell.
- Hirsch, Aron. 2015. "A compositional semantics for wh-ever free relatives." In *Proceedings of Sinn und Bedeutung 20*, edited by Nadine Bade, Polina Berezovskaya, and Anthea Schöller. 341–358.
- Hoekstra, Eric, Bouke Slofstra, and Arjen P. Versloot. 2009. "Changes in the use of the Frisian quantifiers *ea/oait* 'ever' between 1250 and 1800." In *Historical Linguistics 2009: Selected papers from the 19th International Conference on Historical Linguistics*, edited by Ans M.C. van Kemenade, and Nynke de Haas. 171-189.
- Jespersen, Otto. 1917. "Negation in English and other languages." Copenhagen: Høst.
- Jouön, Paul. 1923. "Grammaire de l'hébreu biblique." Rome: Institut biblique pontifical.
- Kadmon, Nirit, and Fred Landman. 1993. "Any." *Linguistics & Philosophy* 16: 353–422.
- Kratzer, Angelika, and Junko Shimoyama. 2002. "Indeterminate Pronouns: The View from Japanese." In *The Proceedings of the Third Tokyo Conference on Psycholinguistics*, edited by Yukio Otsu. Hituzi Syobo, Tokyo. 1–25.
- Krifka, Manfred. 1995. "The semantics and pragmatics of polarity items." *Linguistic Analysis* 25: 1-49.
- Krifka, Manfred. 2006. "Approximate interpretations of number words: A case for strategic communication." In *Cognitive foundations of interpretation*, edited by Gerlof Bouma, Irene Krämer, and Joost Zwarts. Amsterdam: Royal Netherlands Academy of Arts & Sciences. 111–126.
- Kuroda, Shige-yuki . 1965. "Generative grammatical studies in the Japanese language." Doctoral dissertation. Massachusetts Institute of Technology.
- Lahiri, Utpal. 1998. "Focus and negative polarity in Hindi." *Natural Language Semantics* 6: 57–123.
- Lasnik, Peter. 1999. "Pragmatic halos." *Language* 75: 522–551.
- Lehmann, Christian. 1985. "Grammaticalization: Synchronic Variation and Diachronic Change." *Lingua e Stile* 20: 303-318.
- Link, Godehard. 1983. "The logical analysis of plurals and mass terms: A lattice-theoretical approach." In *Meaning, use, and interpretation of language*, edited by Rainer Bäuerle, Christoph Schwarze, and Arnim von Stechow. Berlin: De Gruyter. 302–323.
- Link, Godehard. 1987. "Generalized quantifiers and plurals." In *Generalized quantifiers: Linguistic and logical approaches*, edited by Peter Gärdenfors. Dordrecht: Reidel. 151–180.
- Lipták, Anikó . 2009. "The landscape of correlatives. An empirical and analytical survey." In *Correlatives cross-linguistically*, edited by Anikó Lipták. Amsterdam & Philadelphia: John Benjamins. 1–80.
- Martzlöff, Vincent. 2014. "Some remarks on the prehistory of *omnis* and other Latin pronouns and adjectives meaning 'all' or 'whole'." *Journal of Latin Linguistics* 13: 211-241.

- Matthewson, Lisa. 2001. "Quantification and the Nature of Cross-Linguistic Variation." *Natural Language Semantics* 9: 145–89.
- Menéndez-Benito, Paula. 2010. "On Universal Free Choice Items." *Natural Language Semantics* 18: 33–64.
- Mitrovic, Moreno. 2014. "Morphosyntactic atoms of propositional logic: A philological programme." PhD Dissertation, University of Cambridge.
- Mladenova, Olga. 2009. "The Diachrony of Bulgarian Quantification." *Balkanistica* 22: 89–131.
- Partee, Barbara H. 1987. "Noun-phrase interpretation and type-shifting principles." In *Studies in Discourse Representation Theory and the Theory of Generalized Quantifiers*, edited by Jeroen Groenendijk, Dick de Jongh, and Martin Stokhof. Dordrecht: Foris. 115–143.
- Partee, Barbara H., and Mats Rooth. 1983. "Generalized Quantifiers and Type Ambiguity." In *Meaning, use, and interpretation of language*, edited by Rainer Bäuerle, Christoph Schwarze, and Arnim von Stechow. Berlin: De Gruyter. 361–383.
- Penka, Doris. 2011. "Negative Indefinites." Oxford: Oxford University Press.
- Rawlins, Kyle. 2013. "(Un)conditionals." *Natural Language Semantics* 21: 111–178.
- Roberts, Ian. 1985. "Agreement parameters and the development of English modal auxiliaries." *Natural Language and Linguistic Theory* 3: 21–58.
- Rooth, Mats . 1985. "Association with focus." PhD dissertation. University of Massachusetts, Amherst.
- Rooth, Mats. 1992. "A theory of focus interpretation." *Natural Language Semantics* 1: 75–116.
- Schvarcz, Brigitta R. 2022. "Hárman, soka, mindannyian: A presuppositional analysis to-AN-marked numerals and quantifiers in Hungarian." *Journal of Uralic Linguistics* 1: 181–214.
- Schwarz, Florian. 2013. "Maximality and definite plurals: Experimental evidence." *Proceedings of Sinn und Bedeutung* 17. 509–526.
- Shimoyama, Junko . 2001. "WH-constructions in Japanese." PhD dissertation. University of Massachusetts at Amherst.
- Shimoyama, Junko. 2006. "Indeterminate phrase quantification in Japanese." *Natural Language Semantics* 14: 139–173.
- Simon Eszter, and Bálint Sass. 2012. "Nyelvtchnológia és kulturális örökség, avagy korpuszpépítés ómagyar kódexekből. [Language technology and cultural heritage: Building corpora from Old Hungarian Codices.]" In *Általános Nyelvészeti Tanulmányok [Studies in General Linguistics] XXIV*, edited by István Kenesei, Gábor Prószéky, and Tamás Váradi. Budapest: Akadémiai Kiadó. 243–264.
- Simon, Eszter. 2014. "Corpus building from Old Hungarian codices." In *The Evolution of Functional Left Peripheries in Hungarian Syntax*, edited by Katalin É. Kiss. Oxford: Oxford University Press. 224–236.
- Szabolcsi, Anna. 1986. "Comparative superlatives." *MIT Working Papers in Linguistics* 8: 245–265.
- Szabolcsi, Anna. 1997. "Strategies for scope taking." In *Ways of scope taking*, edited by Anna Szabolcsi. Dordrecht: Springer Netherlands. 109–154.
- Szabolcsi, Anna. 2015. "What Do Quantifier Particles Do?" *Linguistics & Philosophy* 38: 159–204.
- Taylor, Ann, Anthony Warner, Susan Pintzuk, and Frank Beths. 2003. "The York-Toronto-Helsinki parsed corpus of Old English prose."

Winter, Yoad. 2001. “Flexibility principles in Boolean semantics: Coordination, scope and plurality in natural language.” Cambridge, MA: MIT Press.

Xing, Janet Zhiquan. 2015. “Semantic Change and Grammaticalization of the Universal Quantifier měi 每 in Chinese.” In *Space and Quantification in Languages of China*, edited by Dan Xu and Jingqi Fu. Springer. 95-115.

ⁱ My example.

ⁱⁱ Following Berman (1987), Heim (1990) and von Stechow (1994), I assume that A-quantifiers denote second-order relations between sets of situations.

ⁱⁱⁱ The notation and the second pair of sentences are mine.

^{iv} The difference between collectives (such as *naselenie* ‘population’) and aggregates (such as *narod* ‘nation’) is that the former are uncountable (**edno naselenie*, **dve naselenie*) while the latter are countable (*edin narod*, *dva naroda*). Note that Haspelmath (1995) uses ‘aggregate’ in a very different sense, as a label for what we might call plural individuals (including the denotation of plural count nouns).

^v Chesterman (1991) applies this distinction to supplant the singular-plural dichotomy in order to describe the distribution of the indefinite article in English.

^{vi} My notation.

^{vii} “Universal Semantic Cycle” and “Alternative Semantic Cycle” are two different labels referring to the same set of phenomena.

^{viii} Correlatives are discussed, but only in connection with the *vala*-wh-series.

^{ix} In later work, Halm and Bende-Farkas (2024) argued that these are pluractional adverbials and not strictly speaking universal quantifiers.

^x Bende-Farkas (2024) adopts the following definition of ‘floating quantifier’: “an expression that can be interpreted as a quantifier, and which does not form a —minimal — surface constituent, e.g. a DP, with its associate.”